

BEN-GURION UNIVERSITY OF THE NEGEV
FACULTY OF ENGINEERING SCIENCES
DEPARTMENT OF BIOMEDICAL ENGINEERING

**Electroencephalogram data-based analysis of paroxysmal
slow wave events patterns in brain pathologies**

THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE M.Sc DEGREE

By:

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Supervised by:

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October 2025

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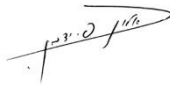
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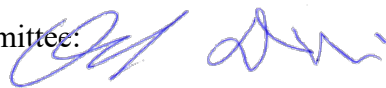
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Abstract

Slowing of brain activity observed in electroencephalography (EEG) recordings is a common phenomenon under resting conditions, such as sleep [1]. However, recent studies have identified a distinct pattern of cortical slowing in patients with neurological disorders, including epilepsy and Alzheimer's disease [2], [3] characterized by transient paroxysmal slow wave events (PSWEs). These events were defined with median power frequency (MPF) less than 6 Hz and duration longer than 5 sec. In this research, I analyzed clinical EEG recordings from the Temple University and Bonn University databases. I systematically characterized the temporal, spectral, and spatial properties of PSWEs in patients with epilepsy, including drug-resistant and drug-responsive subgroups. PSWE features were extracted from each EEG epoch, and machine learning models were trained to classify patients according to epilepsy presence and treatment responsiveness. These results show that PSWE features reliably distinguished epilepsy from non-epilepsy patients and stratified drug-resistant from drug-responsive individuals. Classification models achieved high accuracy across both datasets, demonstrating the robustness and generalizability of PSWE-based biomarkers. Spatial analysis further assessed cortical involvement and multi-focal patterns of PSWE activity. These findings highlight PSWEs as non-invasive EEG markers that capture both the presence and severity of epilepsy. Future studies should validate these results in larger, longitudinal cohorts and explore integration with advanced computational methods, such as deep learning, to improve automated detection and enhance clinical decision-making.

Keywords: Paroxysmal slow wave events (PSWEs); Epilepsy; EEG biomarkers; Drug-resistant epilepsy; Machine learning; Spectral analysis; Classification models; Temporal and spatial EEG features

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Abbreviations

AUC - Area Under the Curve
 ASMs - Anti-Seizure Medications
 BPF - Band-Pass Filter
 DCI - Distribution Concentration Index
 EEG - Electroencephalogram
 EEGLAB - MATLAB toolbox for EEG analysis
 ECoG - Electrocorticography
 FDA - Food and Drug Administration
 FFS - Forward Feature Selection
 FIR - Finite Impulse Response
 FPR - False Positive Rate
 GFI - General Focality Index
 GMM - Gaussian Mixture Model
 HFO - High Frequency Oscillations
 K-Fold CV - K-Fold Cross Validation
 ML - Machine Learning
 MPF - Median Power Frequency
 PCA - Principal Component Analysis
 PSWE / PSWEs - Paroxysmal Slow Wave Event(s)
 ROC - Receiver Operating Characteristic
 SVM - Support Vector Machine
 TLE - Temporal Lobe Epilepsy
 TPR - True Positive Rate
 TUH - Temple University Hospital
 WMED - Weighted Mean Energy Distribution

1 Introduction

Epilepsy is among the most prevalent neurological disorders worldwide, affecting individuals across all age groups, from children to older adults. It is characterized by the occurrence of spontaneous, recurrent seizures that can arise from a variety of underlying causes, including genetic predispositions, structural brain abnormalities, traumatic injuries, tumors, cerebrovascular events, and infections [4]. Globally, it is estimated that approximately 50 million people live with epilepsy, a condition that is often accompanied by cognitive, behavioral, and psychiatric comorbidities, which collectively contribute to a substantial reduction in quality of life [5]. The unpredictability of seizures can have profound effects on daily functioning, educational attainment, employment, and social interactions, highlighting the importance of accurate diagnosis and effective management strategies.

Epilepsy diagnosis is typically based on a combination of clinical evaluation, patient history, and supportive diagnostic tests. Clinicians assess the characteristics of seizures, including their frequency, duration, triggers, and associated neurological symptoms. Medical history often includes reports from patients or witnesses describing seizure episodes, as well as assessment of prior brain injuries, infections, or family history of epilepsy [5],[4],[6]. Among diagnostic tools, electroencephalography (EEG) plays a central role by directly recording the brain's electrical activity and identifying abnormal patterns associated with seizures. EEG can detect interictal epileptiform discharges—brief, abnormal spikes or sharp waves that occur between seizures, as well as ictal activity during a seizure. However, conventional EEG recordings are often limited to short durations in routine clinical practice, which may result in infrequent or subtle epileptic events being missed. Despite these limitations, EEG remains a cornerstone of epilepsy diagnosis, providing essential information about seizure type, localization, and the underlying neural networks involved [7], [8].

Paroxysmal slow wave events (PSWEs) have recently gained attention as promising biomarkers for epilepsy and other neurological conditions. PSWEs are transient episodes of abnormal cortical slowing, reflecting a temporary shift in network activity from normal brain rhythms to low-frequency oscillations. In humans, these events are typically defined as periods in which the median power frequency (MPF) falls below 6 Hz for at least 5 seconds, whereas in rodent models, the threshold is defined as MPF below 5 Hz for more than 10 seconds. Although slow-wave activity can occur

physiologically, for instance during deep sleep or under certain anesthesia conditions, PSWEs have been increasingly recognized as pathophysiological markers of cortical dysfunction [2], [3].

Importantly, PSWEs are not limited to epilepsy. Several studies have reported their presence in other neurological and cognitive disorders, including Parkinson's disease [9], Alzheimer's disease [2], aging [10] and cognitive impairment in patients with obstructive sleep apnea [11]. These findings suggest that PSWEs may reflect broader alterations in cortical network dynamics and could serve as indicators of general neurological vulnerability. In the context of epilepsy, PSWEs have been shown to differ significantly between patients and healthy controls, being both more frequent and of longer duration in individuals with epilepsy [2], [9]. Moreover, PSWEs detected following a first seizure have been proposed as predictive biomarkers for the subsequent development of recurrent seizures, potentially enabling early risk stratification [3]. Specifically, patients who later develop chronic epilepsy tend to exhibit PSWEs with lower MPF and extended duration compared to individuals who experience only a single seizure, highlighting their potential as prognostic indicators [3].

Despite these advances, the characterization of PSWEs remains limited, with most studies focusing primarily on their frequency and duration. Relatively little is known about their detailed temporal evolution, spectral content, or spatial distribution across cortical regions. Furthermore, the mechanisms underlying PSWE generation, their relationship to seizure onset, and their specificity for different types of epilepsy or other neurological conditions remain incompletely understood. A more comprehensive analysis of PSWE characteristics could therefore provide critical insights into the pathophysiology of epilepsy and improve the utility of these events as diagnostic and prognostic tools.

1.1 Research objectives

The present study aims to address this gap by systematically characterizing PSWEs across multiple dimensions, including temporal, spectral, and spatial features. By moving beyond simple measures of frequency and duration, this work seeks to provide a more comprehensive understanding of the underlying dynamics of these transient slow-wave events. Ultimately, the study aspires to enhance the sensitivity and specificity of EEG-based biomarkers, facilitating earlier and more accurate

identification of individuals at risk for developing epilepsy. Such improvements could inform clinical decision-making, guide timely interventions, and support the development of personalized management strategies. In addition to their diagnostic and prognostic potential, a deeper characterization of PSWEs may shed light on the mechanisms underlying abnormal cortical activity, offering new avenues for research into seizure generation, disease progression, and therapeutic monitoring. By elucidating the complex patterns of PSWEs, this study also aims to contribute to the advancement of precision medicine approaches in epilepsy and related neurological disorders, potentially enabling more targeted and effective interventions.

This research was presented at two conferences:

- 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Copenhagen, Denmark, 2025
- Annual Conference of the Israeli League Against Epilepsy (ILAE), 2025

In addition, an article that presents this study was accepted for publication at the IEEE Engineering in Medicine and Biology Conference (EMBC2025):

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2 Literature review

2.1 Epilepsy

According to the International League Against Epilepsy (ILAE) epilepsy is defined as a brain disorder with: " (1) At least two unprovoked (or reflex) seizures occurring >24 h apart; (2) one unprovoked (or reflex) seizure and a probability of further seizures similar to the general recurrence risk (at least 60%) after two unprovoked seizures, occurring over the next 10 years; (3) diagnosis of an epilepsy syndrome" [12]. Epilepsy is one of the most common neurological disorders in children and adults, characterized by the occurrence of spontaneous seizures derived from different etiologies such as genetics, head trauma, tumors, cerebrovascular diseases and infectious [4]. About 50 million people worldwide are epileptic, often suffering from reduced quality of life and neuropsychiatric impairments [5]. The first line treatment for epilepsy are medications, and yet, about 30% of patients are drug-resistant and referred to other treatments such as neurosurgical operation, neuro- stimulation and dietary therapy [6].

Epileptic seizures are defined as transient episodes of abnormal neurological function resulting from excessive, hypersynchronous electrical activity in neuronal networks of the brain. According to the International League Against Epilepsy (ILAE), a seizure is a short-term occurrence of signs and/or symptoms caused by abnormal, excessive, or synchronous neuronal firing [12]. Unlike other transient neurological events, epileptic seizures arise from an intrinsic dysfunction of cortical excitability and synchronization, which disrupts the delicate balance between excitatory and inhibitory signaling in the central nervous system [4]. Clinically, seizures can present with a wide spectrum of manifestations depending on the brain region involved. These may include motor symptoms such as tonic (sustained muscle contraction) or clonic (rhythmic jerking) movements, sensory phenomena like visual flashes or tingling sensations, behavioral arrest, automatisms, or alterations in consciousness [13]. Importantly, seizures are episodic and typically self-limited, but their unpredictable occurrence can cause significant physical harm, psychological burden, and social limitations. They may be triggered spontaneously or, in some cases, provoked by specific factors such as sleep deprivation, flashing lights (reflex seizures), or metabolic disturbances. The wide variability in clinical expression underlines the heterogeneity of epilepsy syndromes and emphasizes the need for accurate classification and individualized treatment approaches [4].

2.2 Electroencephalogram (EEG)

EEG measures the electrical activity of the brain by using recording electrodes that are placed on the scalp and detect the collective excitatory and inhibitory postsynaptic potentials generated by large groups of neurons at a depth of several millimeters, firing synchronously and producing a strong electric field [14]. An EEG signal's amplitude is determined by the level of synchrony of the neurons: Large amplitude signals are produced when a group of neurons fire synchronously, while asynchronous excitation results in a low-amplitude irregular EEG signal. EEG recording can be done using two types of montages, bipolar and unipolar: bipolar recording consists of 2 electrodes located close to each other, one of them positive and one negative, and the potential difference is measured. In unipolar recording the potentials are measured in location of one electrode relative to a second remote reference electrode [1]. Figure 1 presents an example of a seizure lasting roughly 110 seconds, with marked onset and termination. During the seizure, EEG amplitudes are substantially higher than baseline activity. The event shows a dynamic progression, characterized by gradual changes in waveform morphology and dominant frequency. An artifact observed around 73 seconds likely reflects temporary electrode disconnection, probably caused by convulsive movements [7].

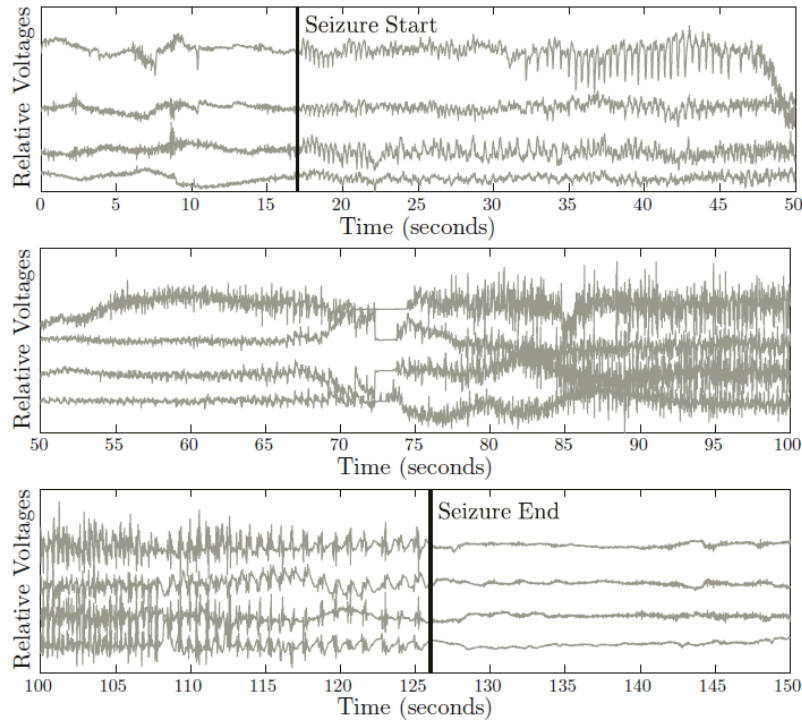


Figure 2.1 Example of an epileptic seizure in EEG [10]. The figure illustrates a typical EEG recording of an epileptic seizure, showing a sudden transition from normal background activity to high-amplitude, rhythmic spike-and-wave discharges.

2.3 EEG-based biomarkers for epileptogenesis

According to the FDA-NIH Joint Leadership Council (2015) [15], a biomarker is a characteristic that is measured as an indicator of normal biologic processes, pathogenic processes, or responses to an exposure or intervention. It includes categories such as susceptibility/risk, diagnostic, monitoring, prognostic predictive, pharmacodynamics/response and safety. A prognostic and diagnostic biomarker for epilepsy is particularly necessary after first seizure or brain insult, when it is uncertainty whether patient will develop epilepsy [16] [17]. Several modalities have been suggested as biomarkers for epileptogenesis, including molecular (plasma HMGB1 and hair cortisol), imaging (brain MRI) and electrophysiology [18]. EEG-based biomarkers, in particular, have been extensively investigated, and the following section will provide a detailed overview of those proposed in previous studies for epilepsy.

2.3.1 Seizures detection

Epileptic seizures, including SE, can be recorded by EEG or ECoG recordings. Detection and characterization of seizure activity (number of seizures/day, seizure duration and total time under seizures) during early days-weeks following first ictal event can serve as a reliable biomarker for the development of the chronic epileptic phase [19].

2.3.2 Theta and gamma waves

Theta waves are brain rhythms occurring in the range of 4-7 Hz, usually associated with drowsiness or certain stages of sleep [1]. In a rats model of temporal lobe epilepsy (TLE), a decrease in theta activity was correlated with cognitive impairments and was suggested as a potential biomarker for epilepsy [20]. In an additional animal study, Milikovsky et al (2017) showed in five different models of post-injury epilepsy that a decrease in theta waves during epileptogenesis is a reliable predictor of epilepsy, with high sensitivity and specificity and that the degree of theta shift may also serve as a prognostic indicator for the duration of the epileptogenic period. Furthermore, a discernible inverse relationship between theta waves and high-gamma activity (30-100 Hz) was also observed [21].

2.3.3 High frequency oscillations

High frequency oscillations (HFO) are fast neuronal oscillations in the range of ~ 100 -600 Hz, that first were recorded experimentally in the hippocampus as part of sharp-wave ripple complexes [22] [23]. A physiological role for HFOs has been suggested in sensory information processing, including learning and memory consolidation [24], planning [25] and decision making [26].

Pathological HFOs (pHFOs), faster ripples (200-600 Hz) are associated with abnormal neuronal network activity in the entorhinal cortex and hippocampus of epileptic patients [27]. Also, increased HFOs were described in the seizure onset zone in patients with focal epilepsy [28], and early detection of pHFOs was associated with an increased rate of spontaneous seizures [29]. However, as fast ripples were also found in normal physiological functions [30] [31], differentiation between epileptiform and physiological HFOs is still a challenge. However, in the present project, since data was collected with sampling rate of 200 Hz, analysis for pHFO was excluded.

2.3.4 Asymmetry

The term brain asymmetry (lateralization) represents structural and functional asymmetries that present in the brain, starting early in development. Physiologically, for example, it is well known today that in most people the left hemisphere is dominant for language and speech, while the right hemisphere is dominant for visuospatial functions [32]. In EEG, asymmetry refers to a difference in activity (including amplitude and/or frequency) between specific regions in the two hemispheres [33]. Benets et al (2018) reported that asymmetry in the background activity of EEG recordings can serve as a reliable biomarker for post-stroke epileptogenesis during the first year after the insult [34].

2.3.5 Sleep stages

A normal sleep cycle consists of muscle relaxation, a decrease in brain activity, and a gradual synchronization of brain waves with occasional bursts of rapid activity called sleep spindles. The brain switches between several sleep states during the night, each characterized by different patterns of brain activity, eye movements, and muscle tone. Sleep stages can be divided into sleep without rapid eye movement (non-REM), and sleep with rapid eye movement (REM) [11]. While during REM sleep, we experience the most vivid dreams which are characterized by low-amplitude synchronous activity in the theta band, among the non-REM stages are drowsiness (stage 1), light sleep (stage

2) and deep sleep (stage 3) that defined by high-amplitude and slow-frequency synchronous activity in delta band. Seizures during sleep can disrupt normal sleep cycle and lead to frequent awakenings, which may result in sleep deprivation and daytime tiredness [35]. Pedro Andrade et al (2017) suggested a relation between sleep stages and post-traumatic epilepsy; They showed that most of spontaneous generalized seizures (92%) occur during the transition from sleep stage 3 to REM. Also, the researchers showed that a reduction in spindle duration and dominant frequency during the transition between sleep stages could be used as a reliable predictor of epileptogenesis with high sensitivity and specificity [36].

2.4 Paroxysmal slow wave events (PSWEs)

Slowing of brain activity observed in electroencephalography (EEG) recordings can be physiological under specific conditions such as deep sleep. However, in a series of studies the BBB Lab has recently described a new, abnormal pattern of cortical slowing called paroxysmal slow wave events (PSWEs) [2], [3]. PSWEs are defined as slow transient events, in which the network switches from apparently normal activity to brief periods of low-frequency activity (in humans, median power frequency (MPF) $<6\text{Hz}$ for more than 5 seconds; In rats, $\text{MPF} < 5\text{Hz}$ for more than 10 seconds) [2], [3].

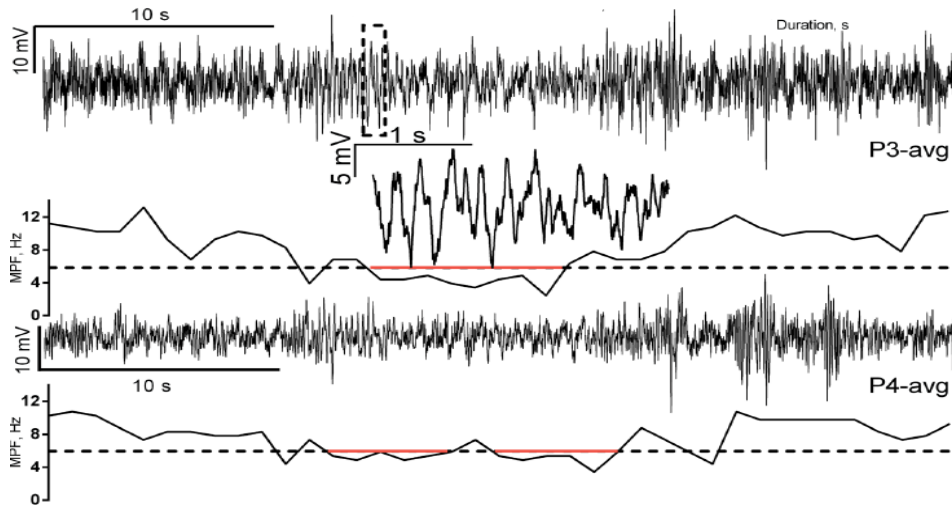


Figure 2.2. Example of a paroxysmal slow wave event (PSWE, in red) in EEG recordings from electrodes P3 and P4, with corresponding median power frequency (MPF) per second [2]. When the MPF drops below 6 Hz for a duration of at least 5 seconds, the event is identified as a PSWE.

PSWEs have emerged as biomarkers for epilepsy and other brain pathologies.

2.4.1 Epilepsy

PSWEs have gained increasing attention as potential electrophysiological biomarkers for epilepsy. Prior studies have shown that PSWEs occur more frequently and exhibit

longer durations in patients with epilepsy compared to controls [2], [9], [37]. Milikovsky et al. (2019) [2] demonstrated that transient paroxysmal slowing in cortical EEG is present in patients with epilepsy and is associated with blood-brain barrier dysfunction, highlighting a mechanistic link between cortical excitability and epileptogenesis. In this research, patients with epilepsy exhibited a significantly higher occurrence of PSWEs compared to both seizure-free patients and healthy controls, indicating their potential as diagnostic biomarkers. Building on these findings, Serlin et al. (2025) [15] reported that PSWEs were more prolonged in patients with epilepsy compared to participants without epilepsy. The effect was most pronounced in focal epilepsy, particularly with temporal lobe involvement, and in patients with drug-resistant epilepsy, indicating an association with treatment refractoriness. Prolonged PSWEs were also observed in patients with poor surgical outcomes. This evidence supports PSWEs as a non-invasive biomarker not only for epilepsy diagnosis but also for assessing disease severity and predicting treatment responsiveness. Another study has shown that the occurrence of PSWEs in early EEG recordings from patients who later reported spontaneous seizures was significantly higher compared to seizure-free patients and healthy controls. PSWEs exhibit lower MPF and longer duration in patients who later reported spontaneous seizures compared to controls, making them a potential predictive factor for epilepsy [3]. Collectively, these studies suggest that PSWEs represent a robust electrophysiological signature of epileptic activity, reflecting both underlying cortical vulnerability and potential clinical outcomes.

2.4.2 Cognitive impairment in patients with obstructive sleep apnea:

PSWEs have emerged as a potential electrophysiological marker of cognitive dysfunction across various neurological conditions. In the context of obstructive sleep apnea (OSA), several studies have highlighted a significant association between PSWEs and cognitive impairment. Specifically, patients with OSA who exhibit mild cognitive impairment (MCI) show a higher frequency of PSWEs compared to cognitively intact OSA patients. This increase is particularly pronounced during REM sleep, suggesting that certain sleep stages may be more vulnerable to abnormal cortical activity. In addition, the occurrence of PSWEs in specific EEG channels during REM sleep has been independently linked to cognitive deficits, indicating that not only the overall PSWE burden but also their spatial distribution may be relevant. These findings underscore the potential utility of PSWEs as a non-invasive biomarker for identifying

OSA patients at risk of cognitive decline and may provide insight into the neurophysiological mechanisms underlying sleep-related cognitive dysfunction [11].

2.4.3 Alzheimer's disease (AD)

In Alzheimer's disease (AD), PSWEs have also been identified as a prominent feature of altered cortical activity. Studies have reported that the occurrence of PSWEs per minute is higher in AD patients compared to age-matched healthy controls. Importantly, the frequency of these events has been shown to correlate with cognitive impairment, as measured by standard assessments such as the Mini-Mental State Examination (MMSE), suggesting a link between PSWEs and the severity of cognitive deficits. In addition to their association with cognitive dysfunction, PSWEs in AD have been linked to underlying pathophysiological changes, including blood-brain barrier dysfunction, which may contribute to the generation of abnormal cortical slow wave activity. These observations support the idea that PSWEs are not only a marker of network instability in AD but may also reflect broader mechanisms of neurodegeneration and disrupted brain homeostasis, emphasizing their potential role as a non-invasive biomarker for monitoring disease progression and cognitive decline [2]

2.4.4 Parkinson's Disease (PD)

In Parkinson's disease, PSWEs appear less frequently compared to other neurodegenerative disorders, yet they may still hold clinical relevance. Studies have reported that, although uncommon, PSWE was significantly correlated with disease duration in PD patients in all brain areas. This relationship suggests that even sporadic occurrences of PSWEs may reflect progressive alterations in cortical network activity as the disease advances. While PSWEs in PD are not as strongly linked to cognitive impairment as in Alzheimer's disease or OSA, their presence may indicate underlying neurophysiological changes associated with disease progression, offering potential insights into the temporal evolution of cortical dysfunction in Parkinson's disease [9].

2.4.5 Aging

In the context of normal aging, PSWEs have also been observed, with their prevalence increasing in older individuals. Research indicates that older participants not only exhibit a higher frequency of PSWEs, but these events also tend to be longer in duration, slower in frequency, and more widespread across the cortex. Moreover, the presence and characteristics of PSWEs have been associated with lower cognitive performance, suggesting that even in healthy aging, these abnormal slow wave events may reflect

subtle changes in neural network dynamics that relate to cognitive decline. These findings highlight that PSWEs are not solely a feature of pathological conditions but may also serve as sensitive markers of age-related alterations in cortical activity and cognitive function [10].

Overall, PSWEs represent a distinct pattern of abnormal cortical slowing that appears across a range of neurological conditions as well as in normal aging. The studies reviewed indicate that PSWEs are more frequent and pronounced in pathological states such as epilepsy, obstructive sleep apnea, and Alzheimer's disease, where they are consistently associated with cognitive impairment or disease progression. In Parkinson's disease, although PSWEs are less common, their occurrence correlates with disease duration, suggesting a link to progressive network dysfunction. Even in healthy aging, PSWEs increase in frequency, duration, and spatial extent, and are associated with lower cognitive performance, highlighting their sensitivity to subtle changes in brain function. Collectively, these findings support the emerging view that PSWEs serve as a non-invasive electrophysiological biomarker of cortical vulnerability, reflecting both disease-specific and age-related alterations in brain networks. Their detection may therefore provide valuable insights into the mechanisms of cognitive decline, disease monitoring, and potentially early diagnosis across diverse neurological and aging-related conditions.

2.5 Machine learning algorithms

Machine learning (ML) [38], [39],[40], is a subfield of artificial intelligence that focuses on the development of algorithms capable of learning patterns and making predictions or decisions from data without explicit programming. ML techniques aim to extract meaningful insights, identify relationships between variables, and enable models to generalize from observed instances to unseen data. These capabilities make ML an essential tool in domains ranging from healthcare and finance to engineering and natural language processing.

A classification algorithm represents a central paradigm in machine learning, designed to assign categorical labels (or “classes”) to data instances based on their observed features. Classification is widely used in applications such as medical diagnosis, fraud detection, image recognition, and many other areas where decisions are categorical in nature. By framing problems as classification tasks, ML models can provide actionable insights and automate decision-making processes.

2.5.1 Supervised vs. unsupervised learning

Supervised learning refers to scenarios where data is labeled, meaning that each instance is associated with a known output or class. The objective is to learn a mapping from inputs to outputs that allows accurate prediction of unseen instances. Common supervised tasks include classification and regression. By contrast, unsupervised learning deals with unlabeled data, where the algorithm must infer underlying structures or patterns without explicit guidance. Unsupervised methods include clustering, which groups similar instances together; dimensionality reduction, which simplifies high-dimensional datasets while retaining key information; and identification of latent associations among features. The primary distinction between these paradigms lies in their objectives: supervised learning optimizes predictive performance using known outcomes, whereas unsupervised learning explores the intrinsic structure of the data to uncover hidden patterns or relationships [38], [39],[40].

2.5.2 Data splitting

To ensure robust evaluation and prevent overfitting, datasets are typically divided into three subsets: training, validation, and testing. The training set is used to fit the model and learn patterns from the data. The validation set guides model selection, parameter tuning, and processes such as feature selection, allowing the model to improve generalization without relying on the test set. The testing set, kept separate throughout

the training and validation phases, provides an unbiased assessment of the final model's performance. A common practice is to allocate approximately 80% of the dataset for training (with a portion reserved for validation) and 20% for testing, while maintaining a balanced distribution of classes across all subsets. This approach ensures that the evaluation reflects the model's expected performance on new, unseen data [40].

2.5.3 Performance evaluation

Evaluating model performance is a critical step in ML, ensuring that models generalize well beyond the training data but also generalize effectively to unseen instances. Several complementary metrics are commonly employed to provide a comprehensive assessment of a model's predictive capabilities. **Accuracy** represents the proportion of instances that are correctly classified, offering a simple overall measure of performance. However, accuracy alone may be misleading in datasets with imbalanced class distributions, where one class dominates.

To address this, **sensitivity** (also known as recall or True Positive Rate) and **specificity** (True Negative Rate) are often used. Sensitivity quantifies the model's ability to correctly identify positive instances, reflecting its effectiveness in detecting the target condition. Specificity, on the other hand, measures the proportion of correctly identified negative instances, indicating how well the model avoids false positives. By considering both sensitivity and specificity, researchers can better understand the trade-offs between different types of errors and make informed decisions depending on the application context.

Graphical tools, such as the **Receiver Operating Characteristic (ROC) curve** (Figure 2.3) and the **Area Under the Curve (AUC)**, complement these scalar metrics by illustrating the model's discriminative ability across varying classification thresholds. Receiver Operator Characteristic (ROC) curve [41], is a graphical representation of the model's performance across various classification thresholds. Each point on the ROC curve corresponds to a specific threshold, with the True Positive Rate (TPR) indicating the proportion of correctly identified positive cases, and the False Positive Rate (FPR) indicating the proportion of negative cases incorrectly classified as positive. The AUC-ROC represents the area under the ROC curve, ranging from 0 to 1, and summarizes the overall effectiveness of the binary classification model, with higher values indicating better discriminative power between classes.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.1)$$

$$\text{Sensitivity (Recall, TPR)} = \frac{TP}{TP + FN} \quad (2.2)$$

$$\text{Specificity (TNR)} = \frac{TN}{TP + FP} \quad (2.3)$$

$$\text{FPR} = 1 - \text{Specificity} = \frac{FP}{TP + FP} \quad (2.4)$$

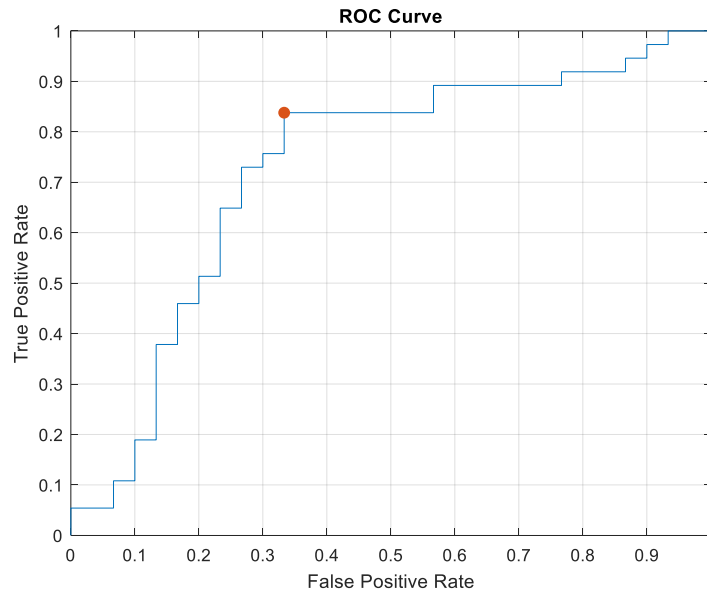


Figure 2.3 Example of ROC curve. The figure presents a Receiver Operating Characteristic (ROC) curve illustrating the trade-off between sensitivity and specificity across different classification

2.5.4 K-Fold cross validation

To robustly assess model performance, **K-fold cross-validation** is commonly employed. In this approach, the dataset is divided into K equally sized subsets, or folds. The model is trained on K-1 folds and validated on the remaining fold. This process is repeated K times, with each fold serving as the validation set once, allowing for a comprehensive evaluation of the model's generalization ability. Cross-validation is particularly useful for smaller datasets, as it reduces potential biases and ensures that each data instance is used for both training and validation.

In some applications, the predictions from the multiple models generated during cross-validation can be combined into an **ensemble or expert system**, which makes final classifications based on majority voting. For example, if more than half of the models

predict a positive class for a given instance, the system classifies it as positive. This aggregation helps improve robustness and reliability of the final predictions.

2.6 Supervised classification algorithms

2.6.1 Support vector machine (SVM)

Support Vector Machine (SVM) [42], is a supervised machine learning algorithm used for both classification and regression tasks. SVM classifies two classes by finding the optimal hyperplane that maximizes the distance (margin) between the classes in an N-dimensional space (Figure 2.4). By maximizing this margin, the algorithm identifies the most effective decision boundary, reducing the risk of misclassification and improving its ability to generalize to new, unseen data. The SVM can use different types of kernels to find this boundary:

- Linear SVM - uses a hyperplane represented by the equation $w \cdot x + b = 0$ (w - weight vector; x - input vector; b - bias).
- Gaussian (RBF) Kernel: Useful for non-linear data, this kernel maps the input features into a higher-dimensional space using a Gaussian function, allowing the SVM to find a more complex decision boundary

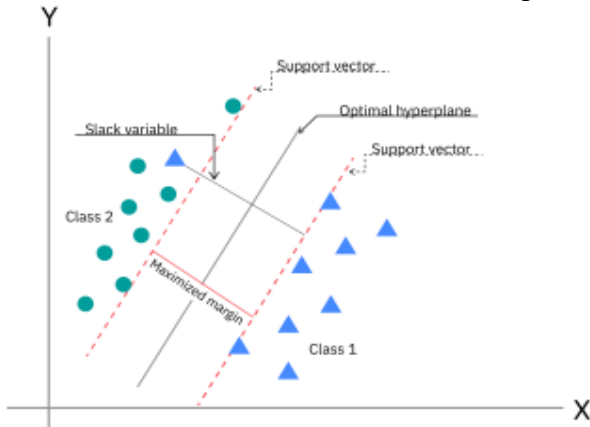


Figure 2.4. SVM algorithm [16].

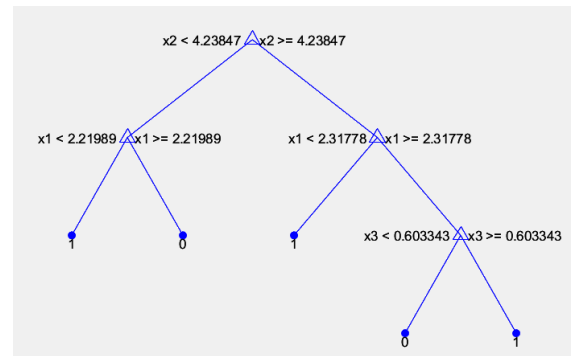


Figure 2.5 Example of decision tree

2.6.2 Decision trees

A decision tree [43], is a non-parametric, supervised machine learning algorithm used for both classification and regression tasks. It employs a divide-and-conquer strategy, performing a greedy search to find the best split points in a top-down, recursive manner until the data is adequately classified. In this structure, each internal node tests a specific feature, each branch represents the result of that test, and each leaf node represents a final prediction or decision. The algorithm aims to split the data based on feature values to form the most homogeneous subgroups, making predictions by following paths from the root node to a leaf node.

2.6.3 Logistic regression

Logistic regression [44], [45], is a statistical method used for modeling the relationship between a categorical dependent variable and one or more independent variables (the features). It predicts the probability of occurrence of an event by fitting data to a logistic curve.

When the observed values of the observation j are:

$$Y_j (1 \text{ for epilepsy and } 0 \text{ for non - epilepsy})$$

$$X_j = (x_{1j}, \dots, x_{Mj}) - \text{when } M \text{ is the number of features}$$

$$\text{if } \pi(X_j) = P(Y_j = 1 | X_j) : \log \left(\frac{\pi(X_j)}{1 - \pi(X_j)} \right) = \beta_0 + \beta_1 x_{1j} + \dots + \beta_M x_{Mj} \quad (2.5)$$

We wish to find the values of $\beta_0, \dots, \beta_M$ which give the "best fit" to the data. The measure of the goodness of fit is the squared error loss (the sum of the squared deviations of the fit from the data points (y_j)), and the best fit is obtained when that function is maximized:

$$\log P(\beta|Y, X) = \sum_{j=1}^N Y_j \cdot \log \pi(X_j) + (1 - Y_j) \cdot \log (1 - \pi(X_j)) \quad (2.6)$$

$$\hat{\beta} = \arg \max \log P(\beta|Y, X) \quad (2.7)$$

2.7 Unsupervised classification algorithms

2.7.1 Independent component analysis (ICA) in EEGLab

Independent Component Analysis [46], [47] is a computational technique used to decompose multivariate signals such as EEG into additive subcomponents. ICA is a specific instance of Blind Source Separation (BSS), a process that aims to extract original source signals from observed mixtures without prior knowledge of the sources themselves or the mixing process. The core principle of ICA lies in identifying components that are statistically independent from one another by optimizing specific criteria, such as maximizing entropy. By leveraging this statistical independence, ICA effectively isolates signals presumed to originate from distinct neural or non-neural processes. This makes ICA a valuable tool in EEG signal processing for both artifact removal and source localization. The ICA algorithm operates under the assumption that brain activity and artifacts arise from anatomically and physiologically distinct processes. Consequently, it assumes that these sources are statistically independent. Each estimated Independent Component (IC) is represented by spatial weights (spatial expression) and its temporal activation. Using these features, the original EEG data can be reconstructed by subtracting each IC's contribution based on its spatial and temporal characteristics, allowing for artifact removal or signal enhancement.

ICA aims to find an unmixing matrix W such that: $S = WX$ where the sources (rows of S) are statistically independent and non-Gaussian. The algorithm seeks to maximize the entropy of the output signals after applying a nonlinear transformation, which promotes independence. Entropy, a measure of uncertainty or information content, is formally defined as:

$$H(\vec{x}) = - \int_{-\infty}^{\infty} p(\vec{x}) \log(p(\vec{x})) d\vec{x} \quad (2.8)$$

The algorithm updates the matrix W iteratively to maximize the output entropy until convergence.

Once ICA decomposition is complete and the ICs have been identified, a crucial step is determining which components should be retained and which should be rejected (e.g., artifacts). EEGLAB provides both manual and automated methods to assist in this process, based on the following criteria:

1. **Dominant Class Probability:** Each IC is assigned a dominant class (e.g., brain, eye, heart, muscle) using classifiers like ICLabel, with an associated probability indicating the likelihood that the source belongs to that class. Components are retained if the dominant class is “brain” and its associated probability exceeds a user-defined threshold.
2. **Residual Variance (RV):** This metric assesses how well a single equivalent dipole model fits the IC's scalp topography. Lower residual variance indicates a better fit, suggesting the source is more likely to be of neural origin. Components with high RV are considered less reliable and may be rejected.
3. **Source Location within the Brain:** By fitting dipoles to each IC and mapping them within a brain volume model, the algorithm estimates their spatial location. Components whose dipoles are localized outside the brain or at an implausible depth are flagged for rejection.

These three criteria are combined in decision-making: components that fail to meet all three thresholds (i.e., low brain probability, high residual variance, and poor localization) are identified for removal, while those that pass are retained for further analysis.

2.7.2 K-means

K-means [48], is an unsupervised machine learning algorithm employed for clustering tasks. It is an iterative algorithm that partitions data into K clusters based on the proximity to their centroids. The process begins with the random initialization of K centroids, followed by the assignment of data points to the nearest centroid. The centroids are then updated as the mean of the points assigned to each cluster. This process is repeated until convergence. Normalization of features is crucial in k-means clustering, as the algorithm relies on distance metrics; normalization ensures that all features contribute equally to the distance calculations, thereby preventing features with larger scales from disproportionately influencing the clustering outcome.

2.7.3 Gaussian mixture model (GMM) + Expectation-Maximization (EM)

Gaussian mixture model (GMM) is machine learning method used to estimate the probability that each data point belongs to a particular cluster. Clustering, an unsupervised learning approach, groups data points based on shared characteristics. GMMs classify data by modeling it as a mixture of K Gaussian distributions, assigning

a probability to each data point indicating its likelihood of belonging to a specific cluster.

The parameters of the Gaussian distributions in GMM are optimized using the Expectation-Maximization (EM) algorithm, an iterative optimization technique for parameter estimation. The EM algorithm consists of two main steps:

- Expectation (E) Step: Calculates the expected value of the log-likelihood function, given the current parameter estimates.
- Maximization (M) Step: Updates the parameter estimates to maximize the expected log-likelihood computed in the E-step.

This iterative process continues until convergence, producing a model that effectively captures the underlying structure of the data [49].

2.8 Unsupervised dimensionality reduction algorithms

To simplify complex models and prevent overfitting, dimensionality reduction algorithms reduce the number of features while retaining essential information. By mapping high-dimensional data into a lower-dimensional representation, these algorithms make the data easier to analyze, visualize, and process. In addition, reducing dimensionality can improve computational efficiency, mitigate the curse of dimensionality, and reveal hidden structures or patterns within the data that may not be apparent in its original form [40].

2.8.1 Feature selection

This method focuses on identifying and retaining the most relevant features from the original dataset while systematically removing those that are irrelevant or redundant. By doing so, feature selection enhances computational efficiency, as it reduces the complexity of the model, leading to faster training and evaluation times. Moreover, this approach maintains the interpretability of the model by preserving the original features, which are crucial for understanding the underlying relationships within the data. By focusing on these key features, the model is better equipped to generalize to new data, thereby improving its overall performance and reliability in real-world applications. Unlike feature extraction, which transforms the features into new representations, feature selection retains the original features while eliminating irrelevant or redundant ones, thus maintaining interpretability and simplifying the model.

2.8.1.1 Correlation-Based filter approach

The Correlation-Based Filter approach [50], is a feature selection method that evaluates the relevance of each feature by measuring its correlation with the target variable while minimizing redundancy between selected features. The goal of this algorithm is to select a subset of features that are most relevant to the target variable, while eliminating highly correlated and redundant features. The algorithm involves the following steps:

1. Calculate the Symmetrical Uncertainty (SU) for each feature and rank them in descending order of relevance. The Symmetrical Uncertainty (SU) coefficient measures the amount of information obtained about one variable through another variable:

$$SU(X, C) = 2 \cdot \frac{MI(X, C)}{H(X) + H(C)} \quad (2.9)$$

X- features vector, C- classes

Mutual information quantifies the amount of information obtained about one variable through the other:

$$MI(X, C) = H(X) + H(C) - H(X, C) \quad (2.10)$$

$$H(X) = - \sum_i P(x_i) \log(P(x_i)) \quad (2.11)$$

Entropy is a measure of the uncertainty in a random variable x:

$$H(X, C) = - \sum_{i,j} P(x_i, c_j) \log(P(x_i, c_j)) \quad (2.12)$$

2. Select the first feature from the ranked list (denoted as X, the most relevant feature) and remove all features that have a high correlation with X. Correlation calculations are based on the Pearson correlation coefficient, which is given by:

$$r(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (2.13)$$

3. Set the next remaining feature in the list as X and repeat step 2 for all remaining features in the list.

2.8.1.2 Forward feature selection

Forward Feature Selection (FFS) [51], is one of the most widely used stepwise feature selection methods for dimensionality reduction in supervised learning tasks. The central idea behind FFS is to identify a compact subset of features that contributes most to the predictive power of a model, while eliminating irrelevant or redundant variables that may introduce noise or increase the risk of overfitting.

The procedure begins with an empty feature set. At each iteration, the algorithm evaluates the performance of models built with each of the remaining candidate features added to the current set. The feature that yields the greatest improvement according to a predefined performance metric (such as accuracy, F1-score, or AUC-ROC) is then included. This iterative process continues until a stopping criterion is reached- for example, achieving a predetermined number of features, reaching a performance plateau, or when no additional feature improves the model beyond a minimal threshold.

One of the key advantages of FFS is its balance between computational efficiency and model interpretability. Unlike exhaustive search, which requires evaluating all possible feature subsets and quickly becomes intractable for high-dimensional data, FFS provides a heuristic yet effective approach that drastically reduces computational complexity. At the same time, because it selects from the original features rather than transforming them (as in feature extraction methods such as PCA), the resulting model remains interpretable, making it easier to relate outcomes back to the original variables.

However, FFS also has certain limitations. Since the algorithm makes greedy decisions at each step, it does not guarantee finding the globally optimal subset of features. The inclusion of a feature early in the process may prevent another subset from being selected later, even if that alternative could yield better performance overall. Furthermore, the computational cost can still be substantial when dealing with very large feature spaces, as each iteration requires retraining and evaluating the model.

2.8.2 Feature extraction

Feature extraction refers to techniques that transform the original features into a new, typically lower-dimensional representation, while preserving the most informative aspects of the data. Unlike feature selection, which removes redundant or irrelevant variables, feature extraction creates new composite features (e.g., via PCA, LDA, or autoencoders) that capture the underlying structure of the dataset. This approach can

improve model performance and reduce computational complexity, but often at the cost of direct interpretability.

2.8.2.1 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) [52], [53], is a widely used dimensionality reduction technique in machine learning that simplifies a large dataset into a smaller set while preserving significant patterns and trends. This approach is particularly useful when dealing with datasets containing numerous features, as it reduces the number of features while retaining as much relevant information as possible. Dimensionality reduction helps mitigate the risk of overfitting and facilitates data exploration and visualization by reducing the complexity of the dataset. PCA ensures that the resulting components are uncorrelated and that the majority of the information from the original variables is compressed into the first few principal components.

The PCA process typically involves the following steps:

- **Standardization:** This step ensures that all features contribute equally to the analysis and avoids biases caused by differences in variable scales. Standardization is achieved mathematically by transforming each value using the formula:

$$\frac{value - mean}{STD} \quad (2.14)$$

- **Calculation of the Covariance matrix:** The covariance matrix represents the

$$cov(x_i, x_j) = E(x_i - E(x_i)) \cdot E(x_j - E(x_j)) \quad (2.15)$$

relationships and correlations between the features of the dataset.

- **Computation of Eigenvectors and Eigenvalues:** Eigenvectors represent the directions of maximum variance (the principal components), while eigenvalues indicate the amount of variance explained by each principal component. The selected eigenvectors are organized into a matrix, known as the feature vector, which contains the principal components to retain.
- **Reorientation of the data:** The original dataset is transformed into a new coordinate system defined by the principal components. This is achieved by multiplying the transpose of the original dataset by the transpose of the feature vector.

3 Goals

The goal of my thesis was to investigate and differentiate the characteristics of PSWEs across epilepsy syndromes, with the objective of improving diagnostic accuracy and deepening the understanding of their clinical relevance.

- **Classification of patient groups:** Investigate the utility of PSWE features for effectively classifying patient groups, focusing on distinguishing between different neurological conditions (such as epilepsy and non-epilepsy states) and exploring their potential diagnostic value.
- **Differentiation of pathological and physiological states:** Explore and categorize different subtypes of PSWEs for differentiation between healthy and diseased brains, and understand the diverse manifestations and clinical implications of PSWEs across various neurological disorders

3.1 The need for research

To date, the characterization of paroxysmal slow wave events (PSWE) has been relatively general, primarily focusing on their frequency of occurrence and duration. This study aims to provide a more comprehensive characterization of these events within both the time and space domains, with the objective of establishing associations with distinct brain pathologies and elucidating their specific features.

In the context of epilepsy, the significance of this research can be categorized into two primary domains: diagnosis and prediction. Epilepsy is diagnosed primarily on clinical grounds, based on seizure semiology and patient history, with EEG and imaging providing supportive evidence. While direct seizure capture on EEG is rare, emerging electrophysiological biomarkers such as PSWEs may offer complementary diagnostic value. Nevertheless, because EEG recordings are not continuous, many seizures remain undocumented, which can contribute to diagnostic uncertainty. The ability to predict individuals at risk of developing epilepsy is equally critical for effective disease management. This is particularly relevant in cases where distinguishing between a single isolated seizure and the onset of chronic epilepsy is challenging, such as after a first seizure or following brain injury. Early prediction and targeted intervention in such scenarios could mitigate the severe consequences associated with epilepsy progression.

4 Methods

4.1 Research data

A cohort of 500 subjects was randomly selected from The TUH EEG Epilepsy Corpus V1.1.0.12 [54], [55]. For each subject, the dataset includes an EEG recording and associated clinical reports. The epilepsy cohort included patients diagnosed with focal, generalized, combined focal and generalized, or undetermined/unclassified epilepsy regardless of EEG findings, while the non-epilepsy cohort included individuals presenting with seizure mimics and normal EEG results. For individuals with multiple recordings, only the first was included in the analysis. Subjects were excluded if they had an ambiguous seizure history, major psychiatric comorbidities, brain lesions, traumatic brain injury, stroke, substance use or incomplete demographic information (age or sex). Additionally, individuals in the non-epilepsy group were excluded if they were taking antiseizure medications (ASMs), except for those using gabapentin, pregabalin, lamotrigine, or topiramate for pain or migraines, and benzodiazepines (e.g., lorazepam, clonazepam) for anxiety. The final dataset comprised 188 subjects diagnosed with epilepsy and 161 subjects without a known diagnosis of epilepsy.

The second dataset, obtained from the University Hospital Bonn, Germany, comprises two groups of epilepsy patients: 70 drug-responsive individuals and 62 drug-resistant individuals. Drug-responsive patients are those whose seizures are effectively controlled with anti-seizure medication (no seizure for at least one year), whereas drug-resistant patients continue to experience seizures despite optimal medical therapy. The latter group includes individuals who achieved seizure freedom for at least one year following surgical intervention. Interictal EEG recordings for all patients in both cohorts were obtained using the standard 10-20 system with 19 electrodes, with a minimum recording duration of 20 minutes. The sampling rate was consistent across all subjects, with 250 Hz for the TUH dataset and 256 Hz for the Bonn dataset.

This study was conducted in alignment with the ethical standards of the Nova Scotia Health Authority and the UKB. Written informed consent was deemed unnecessary, as the research involved a retrospective examination of anonymized patient data, part of which was obtained from an openly accessible public database.

The analysis performed on the data is illustrated in Figure 4.1 and consists of a multi-stage pipeline for EEG-based PSWE characterization. The process begins with signal

preprocessing, including noise removal and filtering, to prepare the EEG recordings for accurate event detection. PSWE events are then identified where the MPF < 6 Hz for at least 5 sec. Following event detection, feature extraction is performed to quantify temporal and spectral characteristics of each event across channels, providing a detailed representation for subsequent analyses. Classification is carried out at two complementary levels. First, patient-level classification aims to provide a diagnostic assessment based on PSWE features, distinguishing epileptic from non-epileptic individuals in the TUH dataset or identifying responders versus non-responders to treatment in the BONN dataset. Second, event-level classification aims to determine whether individual events are pathological or physiological, acknowledging that both types of events may coexist within the same patient. Finally, spatial analysis and image processing are used to examine potential changes in the distribution of PSWEs across scalp regions in different patient groups, allowing the quantification and classification of region-specific involvement. This pipeline integrates signal, feature, and spatial analyses into a comprehensive framework for PSWE characterization.

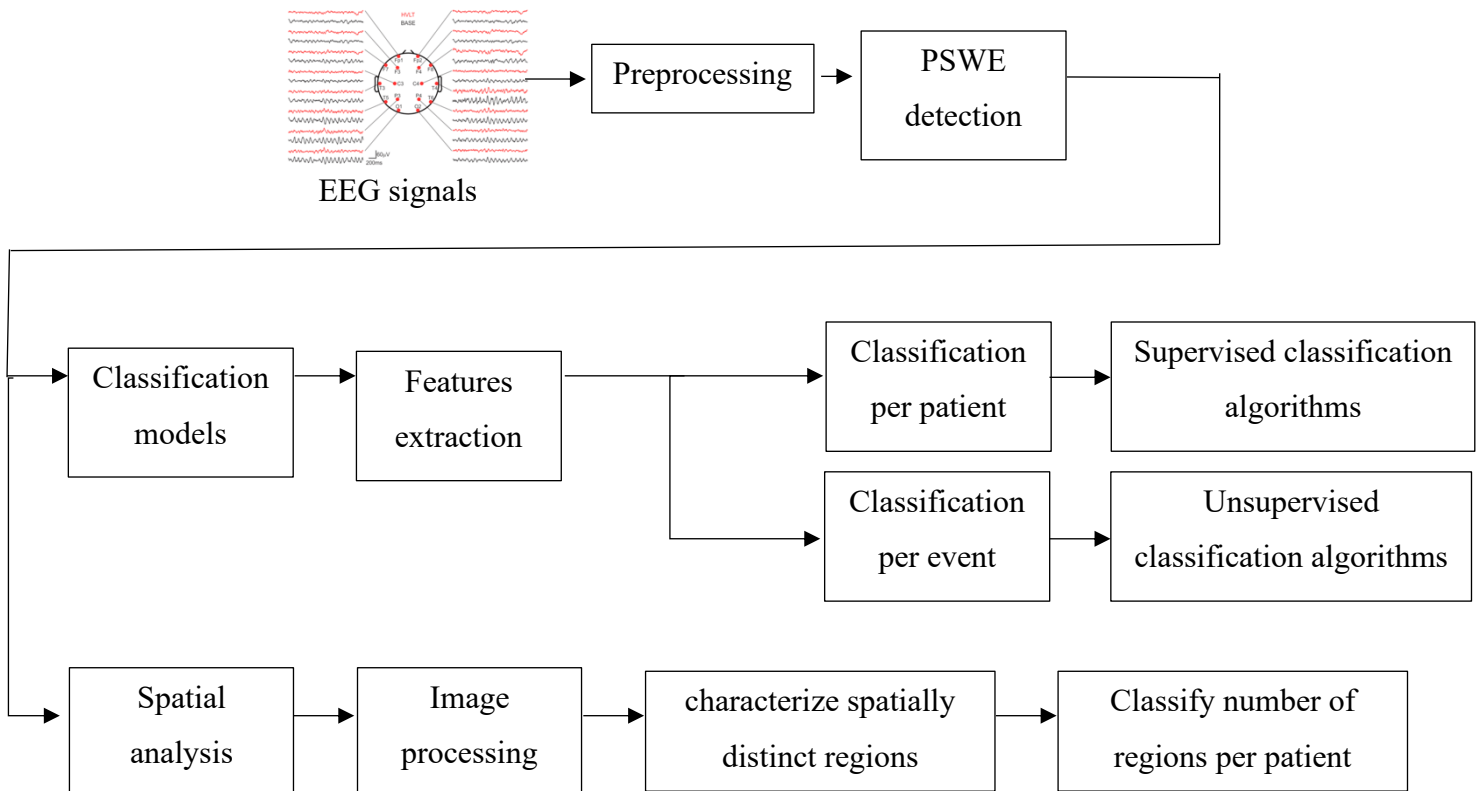


Figure 4.1 Research pipeline

4.2 Preprocessing

The preprocessing was performed based on Makoto's preprocessing pipeline [56] and previous studies [57],[11],[14], and written in MATLAB by a research member in my lab. The preprocessing includes:

BPF 1-45 Hz

Consistent with a previous studies [57],[11],[14] , the PSWEs were characterized after using a band-pass filter (BPF) applied to frequencies ranging from 1 to 45 Hz. Consequently, event detection was performed within the same frequency range to maintain consistency. The BPF was implemented using EEGLAB's default Hamming-windowed sinc FIR filter. There are several reasons to set the lower bound at 1 Hz, including: [56]

- Independent Component Analysis (ICA) can be biased toward high-amplitude components when computed on limited data segments. In the case of EEG, whose power spectral density (PSD) follows a $1/f$ distribution, low-frequency components dominate the variance. As a result, short recording lengths may cause ICA to emphasize slow, high-amplitude activity disproportionately, potentially affecting the separation of sources.
- Low-frequency EEG signals (< 1 Hz) can be contaminated by artifacts such as sweating, introducing spatiotemporal non-stationarity and affecting the performance of ICA.
- The lowest frequency in Event-Related Spectral Perturbation (ERSP) analysis, typically using wavelet or Short-Time Fourier Transform (STFT), is around 3 Hz to ensure sufficient frequency resolution.
- Improved signal-to-noise ratio (SNR).

DC removal

performed by reducing the signal average for each electrode. DC removal in EEG signals is essential to eliminate baseline shifts and low-frequency drifts, as the neutral EEG signal should ideally have no DC component; any detected offset typically arises from artifacts like electrode drift or external interference, potentially distorting neural activity analysis.

Re-reference the data to average

approximation of scalp potentials that is independent of reference location in a location of the head.

Independent component analysis (ICA) in EEGLab [46], [47]

Independent Component Analysis [46], [47] (see Chapter 2.7.1) was performed using the EEGLAB toolbox in MATLAB to decompose multichannel EEG recordings into statistically independent components. The main purpose of this procedure is to identify and isolate both neural and non-neural sources that contribute to the scalp-recorded EEG signals. Neural sources reflect activity originating from different brain regions, while non-neural artifacts typically include eye movements and blinks, cardiac activity, and muscle contractions. By separating these sources, ICA enables the removal of artifacts without compromising the underlying brain signals, thereby improving the signal-to-noise ratio and the reliability of subsequent analyses such as PSWE detection and feature extraction. After decomposition, each independent component is inspected, and components corresponding to artifacts are removed or corrected, while components reflecting genuine neural activity are retained for further analysis. This preprocessing step is crucial for minimizing contamination of EEG signals, ensuring that subsequent analyses accurately reflect brain dynamics rather than extraneous physiological noise.

4.3 Paroxysmal slow wave (PSWEs) detection

PSWEs are defined as slow transient events, in which the network switches from apparently normal activity to brief periods of low-frequency activity (in humans, median power frequency (MPF) $< 6\text{Hz}$ for more than 5 seconds; In rats, $\text{MPF} < 5\text{Hz}$ for more than 10 seconds). The analysis was conducted individually for each electrode, with PSWE events identified as occurrences detected in at least two different electrodes to avoid electrode-related noise/artefacts. To calculate the MPF, I buffered the signal into 1-second windows using MATLAB (according to the sampling rate). For each window, I calculated the power spectrum of the signal, sorted the power spectrum values in ascending order, found the cumulative sum of the sorted values, and determined the frequency at which the cumulative sum crossed half of the total power (i.e., 50% power).

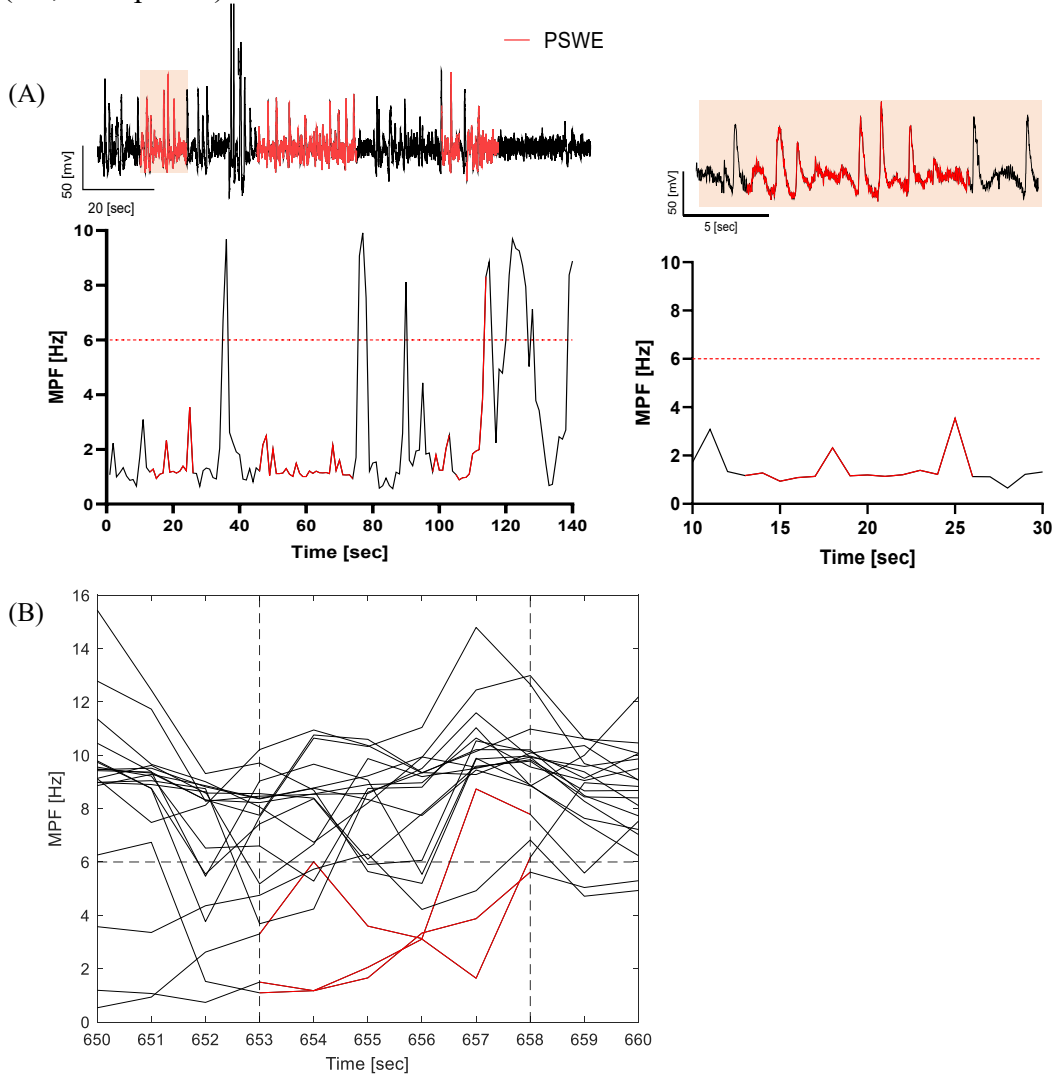


Figure 4.2 Example of a PSWE event. (A) Left: EEG signal from a single electrode and the corresponding MPF per second graph. Right: Zoomed-in view of the first detected PSWE event. (B) Red-marked channels (2, 11, 12) are identified as involved in PSWE from 653 to 658 seconds.

4.4 Features calculation

4.4.1 PSWE features

Based on previous studies, 39 PSWE features have been calculated [3] [58] [59]. The code segments the data into epochs of 2 sec (according to the sampling rate) with a 1 sec overlap. The size was chosen based on previous studies [58], so that it would allow local stationary assumptions and satisfactory temporal and frequency resolution. Let $\mathbf{x} = \{x_1, x_2, \dots, x_N\}$ denote the signal of a PSWE recorded from a single electrode. For every PSWE epoch, features were computed separately for each electrode based on $\mathbf{x}$ and then averaged across electrodes and across patients. The calculated PSWE features are:

1. Energy of the signal: the energy of a signal is a measure of its total power or strength over a given period of time.

$$E = \frac{1}{N} \sum_{n=1}^N x_n^2 \quad (4.1)$$

2. Maximal amplitude: the absolute highest peak – indicates the presence of high amplitude discharge.

$$MA = \max(|x_n|) \quad (4.2)$$

3. Amplitude integral: corresponding to local excitability and synchronicity.

$$AI = \frac{x_1}{2} + \sum_{n=2}^{N-1} x_n + \frac{x_N}{2} \quad (4.3)$$

Amplitude distribution features

4. Standard deviation.

$$\sigma = \sqrt{\frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^2} \quad (4.4)$$

5. Skewness: measures the degree of asymmetry in the amplitude distribution.

$$S = \frac{\frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^3}{\left[\frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^2 \right]^{\frac{3}{2}}} \quad (4.5)$$

6. Kurtosis: describes the relative peakedness or flatness of amplitude distribution.

$$K = \frac{\frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^4}{\left[\frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^2 \right]^2} \quad (4.6)$$

7. Curve length: sensitive to changes in amplitude and frequency.

$$CL = \sum_{n=2}^N |x_n - x_{n-1}| \quad (4.7)$$

8. Harmonicity: estimates the proportion of harmonic components in the power

$$H = \max\{\Gamma\} : \Gamma_m = \sum_{n=0}^{N-m-1} x_n \cdot x_{n+m} \quad (4.8)$$

spectrum. Design to identify the presence of rhythmic synchronized patterns.

where $m = 0, 1, \dots, N - 1$

9. Zero crossing: refers to the points in a signal where the amplitude changes its sign, crossing the zero axis. Zero crossing rate is an estimation of instantaneous frequency, meaning measure of how often these transitions occur within a signal over a specified period of time.

$$ZC = \sum_{n=2}^N (x_n \cdot x_{n-1} \leq 0) \quad (4.9)$$

10. Fractal dimension: a curve is called fractal if it displays self-similarity and can be split into portions, each of which is a scale-reduced copy of the whole. The fractal dimension of a curve indicates the degree of its complexity.

11. Peak

$$P = \frac{\max |x_n|}{\sigma} \quad (4.10)$$

non-linear energy features

$$NE(i) = x_i^2 - x_{i-1} \cdot x_{i+1} \quad (4.11)$$

Estimate both the amplitude envelope and the instantaneous frequency of broad band signals with varying amplitude and frequency components [59].

12-15. mean NE, STD NE, Kurtosis NE, Skewness NE.

Frequency domain feature

16-20. STD FFT, Kurtosis FFT, Skewness FFT, median FFT, sum FFT.

Relative power in brain waves: Relative power refers to the proportion of power within a specific frequency band relative to the total power across all frequency bands in a signal. It provides valuable insights into the relative contribution of different frequency components within the signal. For frequency range of $[a, b]$, the relative power:

$$RP = \frac{1}{FT} \cdot \sum_{j=a}^b f_j ; FT = \sum_{\text{all freq}} f_j \quad (4.12)$$

21. Relative power in delta: 0.5-3 Hz.

22. Relative power in theta: 3-8 Hz.

23. Relative power in alpha: 8-12 Hz.

24. Relative power in beta: 12-20 Hz.

25. Relative power in low gamma: 20-30 Hz.

26. Relative power in high gamma: 30-100 Hz.

Other features per patient:

27. Number of PSWE

28. Mean number of channels that pick up PSWEs - calculate the mean number of channels that pick up the events. It offers an indication of the spatial distribution or involvement of channels in capturing PSWEs within the patient's recordings.

29-30. Max, min num of PSWE per electrode

31. Occurrence per minute – the time of PSWEs as percentage of the total time of the signal.

32. Mean MPF – PSWEs defined when the MPF is lower than 6 Hz for at least 5 sec. This feature looks for the mean of the MPF of specific event (ranges between 1 to 6 Hz).

33. Time in PSWE

34. Mean of event duration – This feature calculates the average duration of PSWEs for a specific patient, providing insight into the typical length of these events within the patient's recordings.

35. Median duration

36-39. Mean MPF per segment, STD of MPF, median MPF, highest MPF

4.4.2 Normalization

As can be seen, the occurrence of PSWEs is normalized with the total time of the recording to prevent recording duration from influencing the results. Additionally, the relative powers are normalized with the total energy across all frequencies. This helps to account for individual differences in overall signal amplitude. By normalizing, I am essentially standardizing the power values to be comparable across different signals or recording sessions. This normalization step is particularly important in comparative analyses where the focus is on the relative contribution of different frequency bands rather than the absolute power values.

I chose not to normalize the other features for several reasons. Firstly, if the features already have similar scales or if the model tolerates variations in feature magnitudes, normalization may not be needed (e.g., all patients have the same 19 channels (or excluded from the analysis in a few who do not), so the the number of channels with PSWEs are out of 19). Moreover, normalization could potentially distort natural feature relationships, especially if they naturally fall within a specific range. Additionally, in some cases, normalization might add unnecessary complexity without significantly improving model performance or preserving information. Lastly, by avoiding normalization, the interpretability of the model's coefficients remains clearer, reflecting the direct impact of each feature on the outcome without scaling adjustments.

Classification separates into per patient and per event:

4.5 Classification per patient

The classification of patient groups seeks to identify differences in PSWE across patient cohorts to enhance our understanding of specific neurological conditions (e.g., elucidating the mechanisms of PSWE in epilepsy to deepen insight into the disorder). Furthermore, this study investigates the utility of PSWE features in accurately classifying patient groups, with a focus on distinguishing between different neurological conditions (such as epileptic and non-epileptic states) and assessing their potential diagnostic value.

The methods involved in this part of the study include:

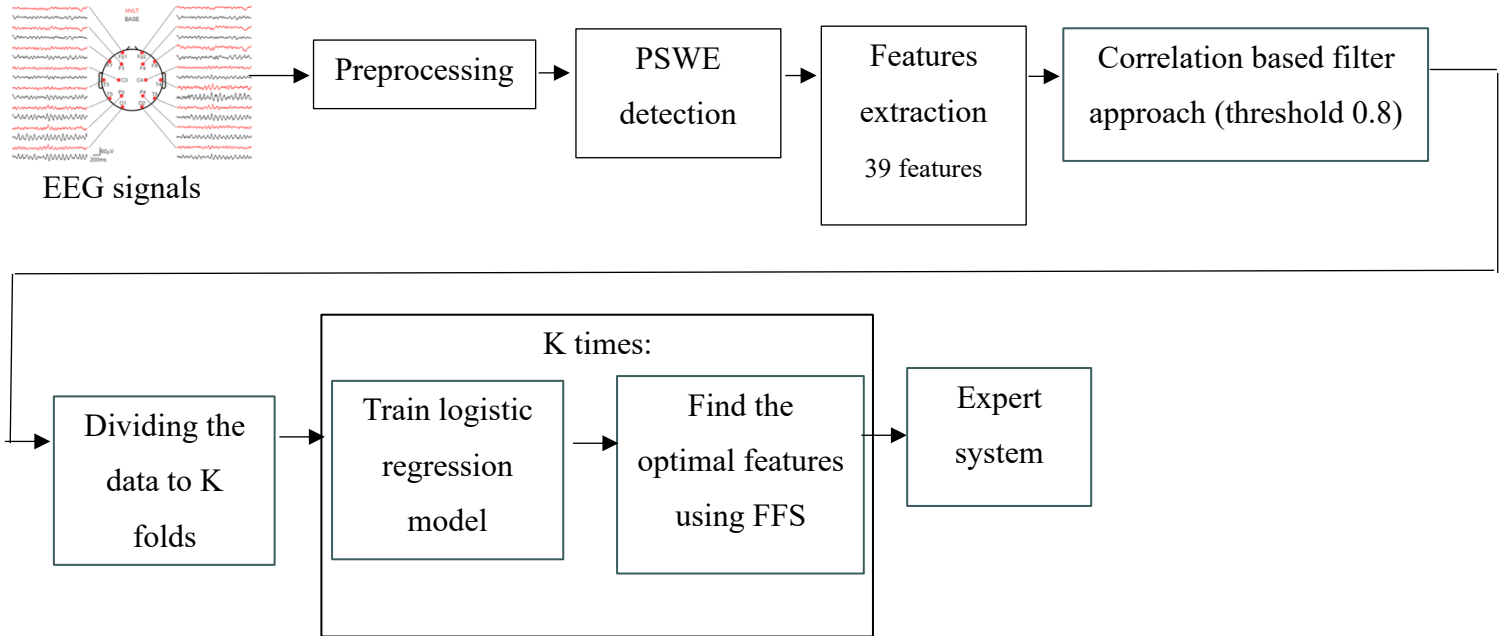


Figure 4.3. Classification per patient pipeline

4.5.1 Feature selection

To reduce model complexity and prevent overfitting, feature selection was applied to identify the most informative features while removing irrelevant or redundant ones. This process improves computational efficiency by reducing the dimensionality of the dataset, accelerates model training and evaluation, and preserves interpretability, as the selected features retain their original meaning, which is essential for understanding the underlying relationships within the data. Two complementary feature selection approaches were employed in this study.

First, a correlation-based filter approach was applied to rank features according to their relevance to the target variable while minimizing redundancy between features. Features were evaluated using Symmetrical Uncertainty, and highly correlated features (Pearson correlation > 0.9) were removed iteratively, resulting in a subset of features that maximally capture relevant information.

Second, forward feature selection (FFS) combined with K-fold cross-validation was used to incrementally construct a feature set that optimizes model performance. Starting with an empty feature set, features were added iteratively, and at each step, model performance was evaluated using the area under the ROC curve (AUC-ROC) as the predefined criterion. The process continued until a stopping condition was met, defined as achieving a minimal improvement of 0.01 in the AUC-ROC relative to the previous model. The final optimal subset balances the number of features with classification performance, preserving interpretability by retaining the original features.

Classification was performed using three supervised algorithms: Support Vector Machines (SVM), logistic regression, and decision tree, and their performance was systematically compared. The dataset was split into training, validation, and testing sets, where the training set was used to fit the models, the validation set guided parameter tuning and feature selection, and the testing set, kept separate throughout, evaluated final model performance. To further ensure robust evaluation, 5-fold cross-validation was applied, with each fold serving once as the validation set while the remaining folds were used for training. The results from all folds were aggregated into an expert system, where final classification of each instance was determined by majority voting across the cross-validated models. This framework ensures that the selected features and classifiers provide both generalizable and interpretable models for PSWE-based EEG analysis.

4.6 Classification per event

Classification of events refers to distinguishing between pathological and physiological states. This involves exploring and categorizing various subtypes of PSWEs to differentiate between healthy and diseased brain activity, as well as understanding the diverse manifestations and clinical implications of PSWEs across a range of neurological disorders. Since there is no established clinical classification for pathologic and physiologic events, we will use unsupervised classifiers to explore and validate this classification.

The methods involved in this part of the study include:

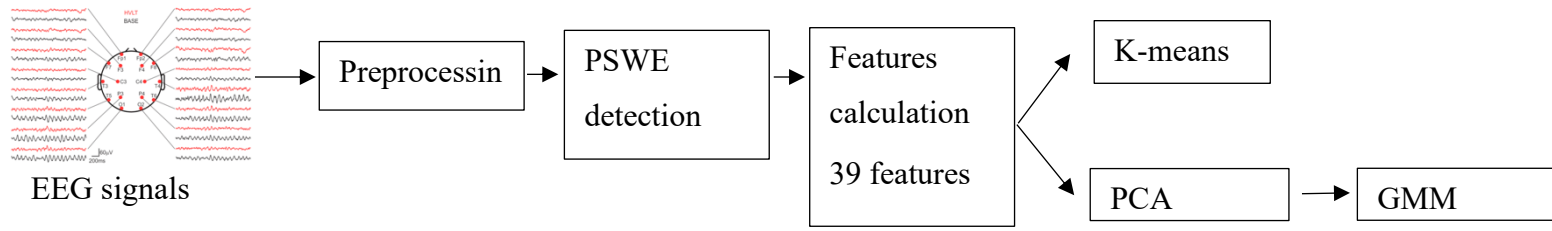


Figure 4.4. Classification per patient pipeline

4.6.1 K-means

K-means clustering was applied to group PSWE events into two clusters, corresponding to pathological and physiological events. Prior to clustering, all features were normalized to ensure equal contribution in the distance-based clustering process. The clustering results were evaluated to confirm the separation of events into the two categories and to assess the consistency and interpretability of the identified clusters.

4.6.2 PCA+GMM

Principal Component Analysis (PCA) was first applied to reduce the dimensionality of the PSWE feature set while retaining the most informative components. The reduced dataset was then clustered using a Gaussian Mixture Model, with the number of clusters set to two, representing pathological and physiological PSWEs. The GMM assigned probabilistic cluster memberships to each event, and the resulting classification was analyzed to validate the distinction between the two event types and to explore the underlying structure of PSWE subtypes.

4.7 Spatial analysis

Spatial analysis aims to identify differences in the spatial distribution of PSWEs across patients' scalp. The analysis focuses on finding the apparent sources of events using scalp topographic map using EEGLAB, a 2D visualization of how data (e.g., voltage or power) is spatially distributed across the scalp (Figure 4.5).

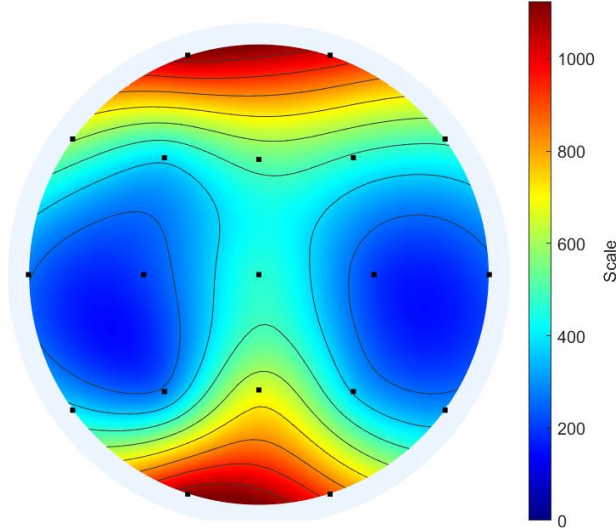


Figure 4.5 Example of a scalp topographic map. Blue indicates regions with no events (value = 0), while red represents the areas with the highest data values.

Using image processing algorithms, the goal is to identify how many separate regions exist and determine their centers. The data used to represent these regions is the normalized energy of the PSWEs, defined as:

$$E_i = \sum_{n=1}^{N_i} x_n^2 ; E_{i,norm} = \frac{\sum_{n=1}^{N_i} x_n^2}{\sum_{n=1}^N x_n^2} \quad (4.13)$$

Where N_i is the number of time points in the i PSWE, and N is the total number of time points in the entire signal.

Next, the **Weighted Mean (WM)** is calculated for each electrode separately, based on the PSWE events occurring at that electrode and their durations D_i . A weighted mean is a type of average in which each data point contributes proportionally to its assigned weight. Unlike a simple arithmetic mean, which treats all

events equally, a weighted mean emphasizes values with greater relevance- in this case, longer-duration events are given more influence. The WM defined as:

$$WM = \frac{\sum_{i=1}^{num\ of\ events} E_{i,norm} \cdot D_i}{\sum_{i=1}^{num\ of\ events} D_i} \quad (4.14)$$

To identify and characterize spatially distinct regions of PSWE activity, an image processing pipeline was developed and applied to scalp topographic maps representing the WM values. The analysis proceeded in five main stages:

1. **Image Preprocessing:** The input gray-level image represented the WM, where higher pixel intensity corresponds to higher values. The image was first rescaled to a [0, 1] range to ensure uniform contrast. A 5×5 filter was then applied to remove isolated low-power (dark) regions and enhance relevant signal areas.
2. **Local Maxima Detection:** Local maxima were identified using MATLAB's 'imregionalmax' function to pinpoint potential centers of high-power clusters. Two thresholds were applied to refine these detections:
 - A proximity threshold removed peaks that were closer than the 20th percentile of the inter-electrode distance distribution (168.238 units), avoiding redundant detections.
 - An intensity threshold excluded peaks with pixel values below the 80th percentile, retaining only prominent high-power regions. Using the 80th percentile provides a robust threshold that consistently captures high-intensity regions without being skewed by outliers or noise, unlike the mean or maximum.
3. **Clustering via GMM:** The retained local maxima served as 'initial guess' for clustering using Gaussian Mixture Model [49]. The algorithm operated only on regions where the pixel intensity exceeded the 80th percentile, using both spatial coordinates and intensity as input features. This allowed for the separation of distinct high-activity regions based on both their location and strength.
4. **Center of Mass Calculation:** For each identified cluster, the center of mass was computed using MATLAB's regionprops. To emphasize higher energy contributions, pixel values were squared prior to this calculation, increasing the influence of strong signal regions on the resulting center positions.

For a labeled region R with pixel coordinates (x_i, y_i) and corresponding intensities $I(x_i, y_i)$ the center of mass is calculated as:

$$\bar{x} = \frac{\sum_{i \in R} x_i \cdot I(x_i, y_i)}{\sum_{i \in R} I(x_i, y_i)}, \bar{y} = \frac{\sum_{i \in R} y_i \cdot I(x_i, y_i)}{\sum_{i \in R} I(x_i, y_i)} \quad (4.15)$$

5. **Merging of Nearby Centers:** As a final refinement, cluster centers that were closer than the predefined proximity threshold were merged, ensuring that only spatially distinct clusters were retained.

This approach allowed for reliable identification of spatial PSWE patterns and their dominant centers, supporting the analysis of inter-patient variability in event localization and distribution. Figure 4.6 shows example of algorithm steps, and figure 4.7 shows examples of center of mass detection (black points).

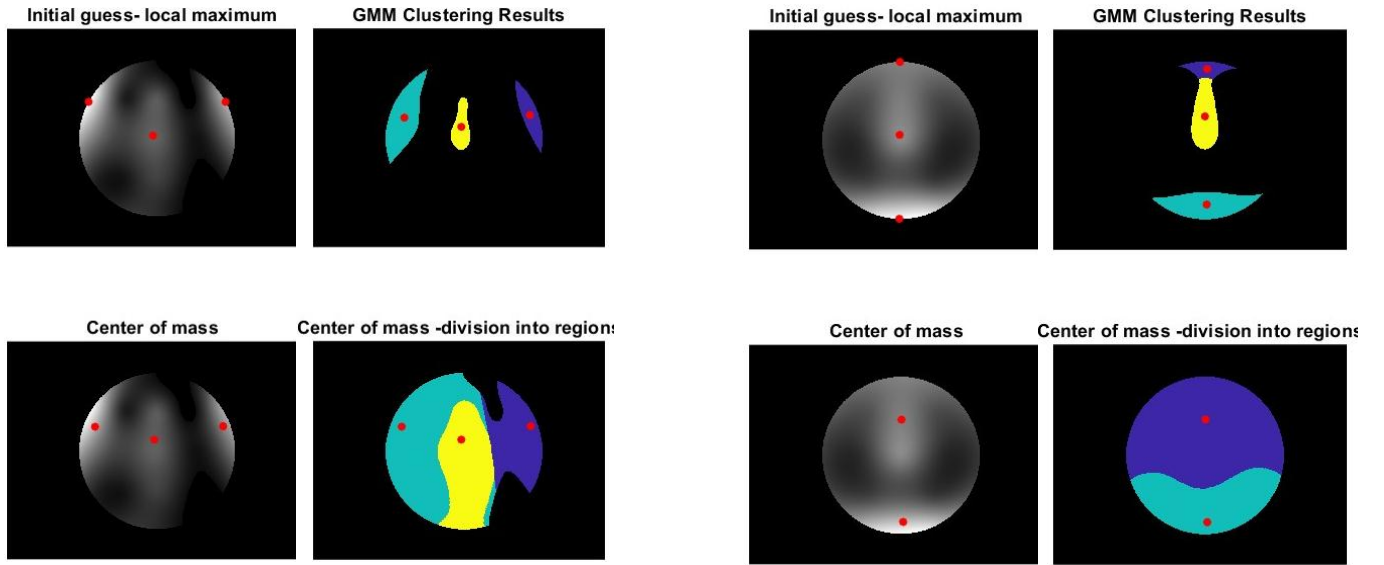


Figure 4.6 Examples of a scalp topographic map and algorithm steps results (red points) for one patient.

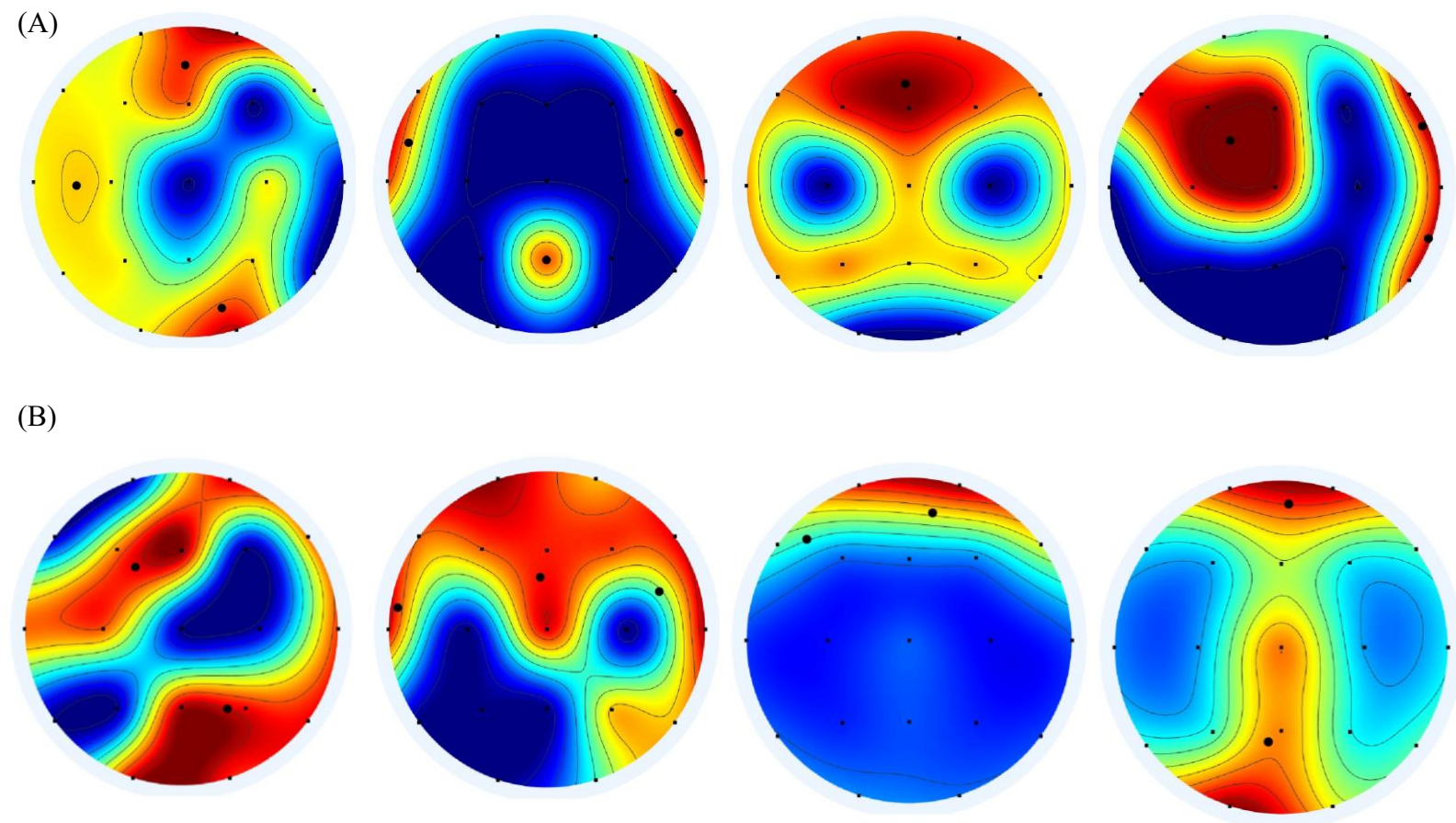


Figure 4.7 Representative examples of center of mass detection across different patients, indicated by black circular markers. (A) Accurate detections. (B) Near-miss detections, with slight deviations from the expected locations.

To compare differences between patient groups, I used the detected regions and their centers to classify patients into four categories: Diffuse, Focal, Focal with other centers, and Multi-Focal.

- Diffuse - widely distributed active regions.
- Focal - a single clearly defined region with high energy, without other significant areas.
- Focal with other centers - one dominant region accompanied by other less significant areas.
- Multi-focal - multiple distinct and significant regions with high energy.

Step 1: Diffusion patients

In order to define whether Weighted Mean (WM) distribution is diffuse in the brain, I combine 3 statistics measurements: mean, contrast and skewness.

1. Mean: Checks the average activity level. High mean means that most of the brain is active at a high level. Since it's grayscale image ranging between [0,1] (with 0 representing black and 1 representing white), the threshold of 80% that means most pixels are close to white.
2. Skewness quantifies the degree of asymmetry of a distribution relative to a perfectly symmetric, bell-shaped (normal) distribution. It indicates whether there is a bias in the data, such as a small, strong focal concentration or a predominance of high values. A skewness value close to zero indicates that the distribution is approximately symmetric, meaning there is no pronounced “tail” of isolated extreme values. In practice, skewness values within the range $[-0.5, +0.5]$ are often considered indicative of a nearly symmetric distribution [60]. Skewness calculates by:

$$Skewness = \frac{\sqrt{n(n-1)}}{n-2} \cdot \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left(\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right)^{\frac{3}{2}}} \quad (4.16)$$

Where n is the number of data points, x_i represents the pixel values and $\bar{x}$ is the sample mean.

An image is classified as '**Diffuse**' if its mean value exceeds 0.8, indicating high activity, and the absolute value of its skewness is less than 0.5, reflecting a near-symmetric WM distribution.

Step 2: Patients with a single detected region

For patients in whom only one region is detected, and who are not classified as 'Diffuse', the image is categorized as '**Focal**'. This implies that activation is confined to a single localized region, with no additional areas identified.

Step 3: Patients with multiple detected regions

Patients with more than one detected region who are not classified as 'Diffuse' can be further categorized as either '**Multi-focal**' or '**Focal with other centers**'. To determine the appropriate classification, I computed three specific metrics:

1. Dominant cluster index (DCI):

$$DCI = \frac{\max_k n_k}{\sum_{k=1}^K n_k} \quad (4.17)$$

Where n_k is the size (number of pixels) of cluster k , and K is the total number of clusters. DCI quantifies how dominant the largest cluster is within a patient's detected regions. A DCI value close to 1 indicates the presence of a single dominant cluster, whereas a value near 0 suggests a more balanced distribution across multiple clusters.

2. Normalized average distance between centroids (D):

$$D = \frac{\text{mean distance between centroids}}{2R} \quad (4.18)$$

The average pairwise distance between all cluster centroids is calculated and then normalized by the radius of the brain image. This measure reflects the spatial distribution of the clusters - a larger D indicates that the clusters are more widely spread across the brain.

3. Weighted max energy differences (WMED):

Each cluster is assigned a normalized energy score, defined as the ratio between the maximum energy within the cluster and the maximum energy across the entire brain area:

$$score_k = \frac{E_k}{\max(E)} \quad (4.19)$$

Then, WMED calculated as the difference between the two highest normalized energy scores:

$$WMED = score_1 - score_2 \quad (4.20)$$

This measure captures the energy gap between the dominant and the second most active cluster, reflecting how dominant the leading cluster is. A high WMED (close to 1) indicates that one cluster stands out significantly.

To integrate these measures into a clustering decision, I applied the k-means algorithm on the three selected features, with $k=2$ corresponding to the classes 'Focal with others' or 'Multi-Focal'. Figure 4.8 illustrates the separation between the two clusters.

To validate the clustering results, I manually labeled 156 patients from the TUH dataset based on their imaging as either 'Focal with others' or 'Multi-Focal'. Comparing these ground-truth labels with the k-means assignments, I found that 144 out of 156 patients (92.3%) were correctly classified.

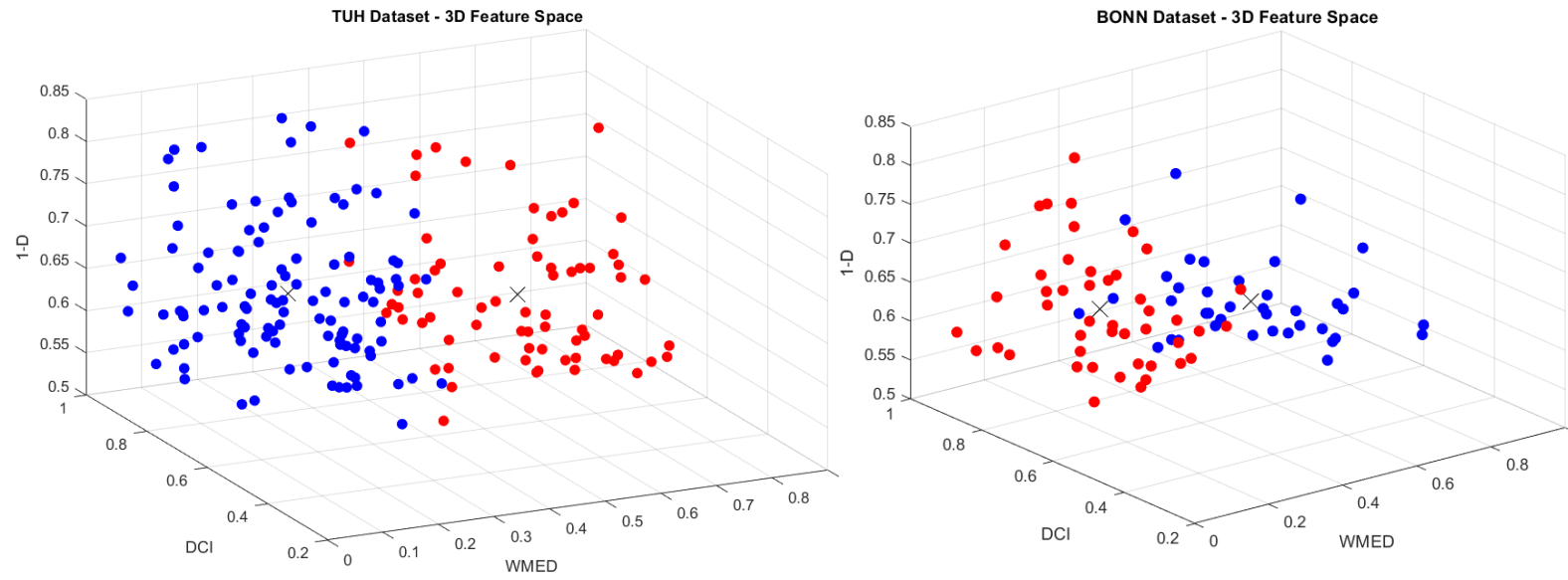
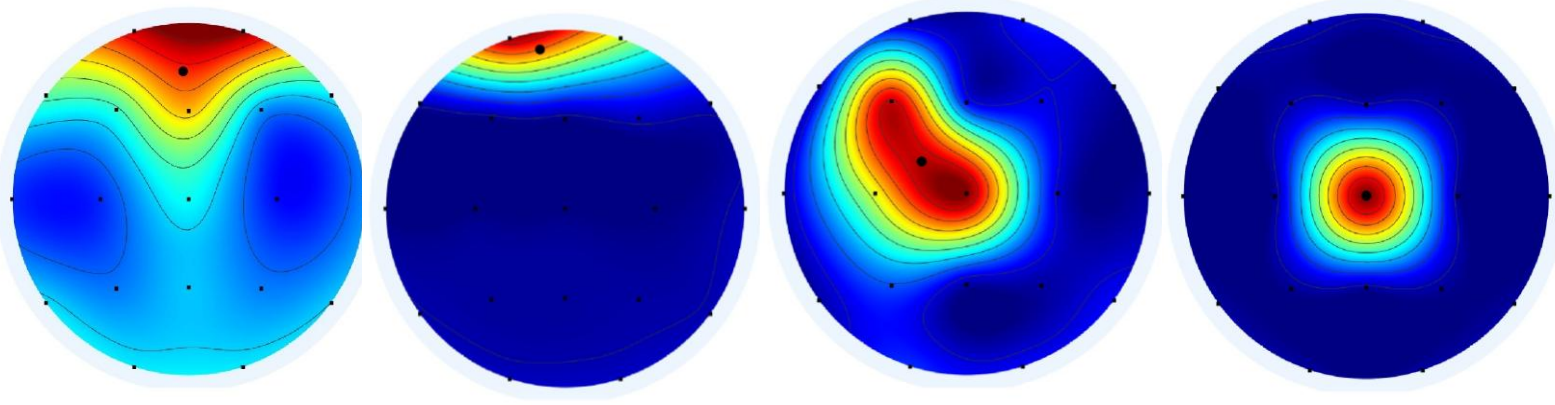
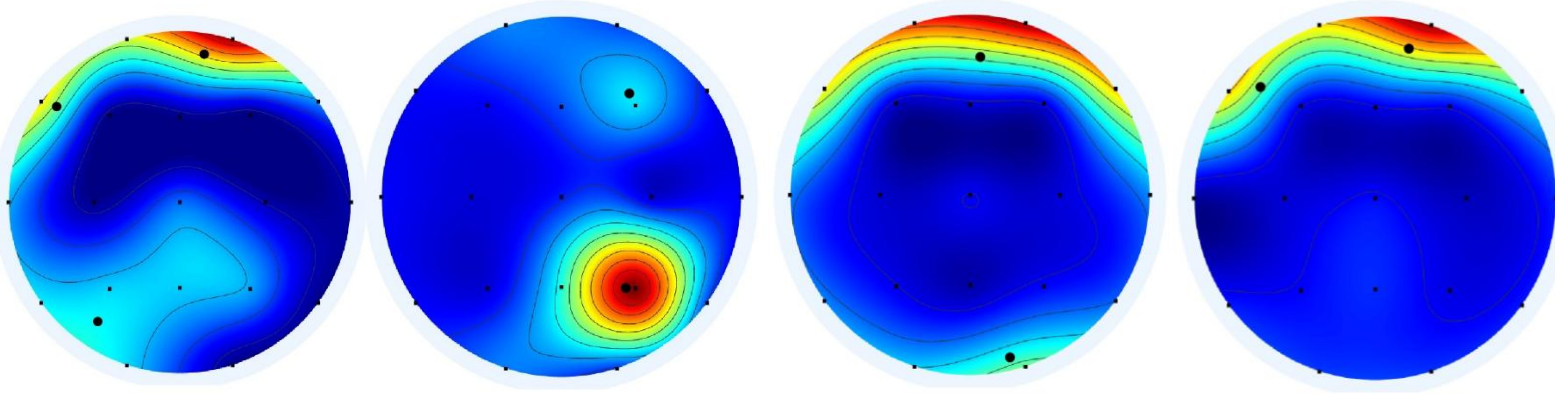


Figure 4.8 K-means algorithm. Left: TUH dataset; Right: BONN dataset.

(A) Focal



(B) Focal with other centers



(C) Multi-focal

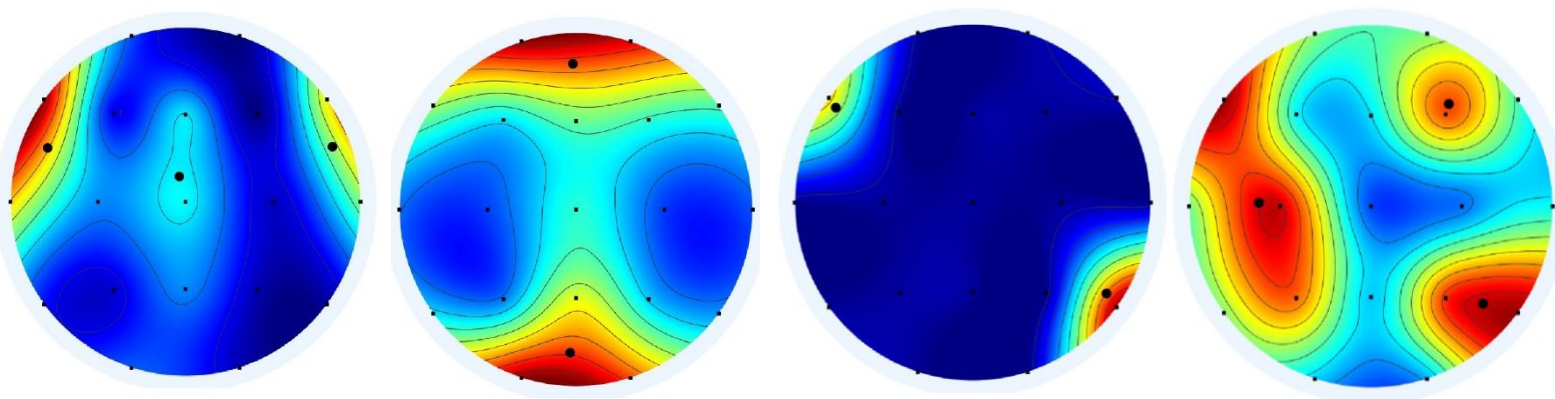


Figure 4.9 Representative examples of the classification. The figure presents representative examples illustrating the differences between the three classification categories: (A) Focal: activity localized to a single scalp region, (B) Focal with other centers: focal activity accompanied by secondary activation in additional regions, and (C) Multi-focal: multiple independent foci of abnormal activity across distinct scalp areas.

5 Results

5.1 Data properties

5.1.1 TUH data

The 'TUH' dataset (see page 11) includes 188 epilepsy patients and 161 non-epilepsy patients. The epilepsy cohort consists of 121 females and 67 males, with an average age of 43.61 ± 14.78 years, while the non-epilepsy cohort includes 101 females and 60 males, with an average age of 43.05 ± 14.78 years. Two patients from each group who exhibited persistent slowing with PSWE activity present in 99% of their EEG recording duration, were excluded from the analysis (186 epilepsy and 159 non-epilepsy patients remaining), as such a high proportion was considered indicative of noise rather than meaningful signal.. On average, each patient exhibited 32.66 ± 17.35 PSWE events.

5.1.2 BONN data

The 'BONN' dataset contains EEGs from 132 patients with known epilepsy who were followed at the Bonn University epilepsy clinic [37]. Out of these 132 patients 70 patients were "drug-responsive" (i.e. free of seizures for at least one year) and 62 were "drug-resistant" patients (i.e. continued to report recurrent seizures despite a combination of anti-seizure medications). The drug-responsive group has 30 females and 40 males, while the drug-resistant group consists of 32 females and 30 males. All drug resistance patients were diagnosed with focal epilepsy and were operated with the goal to remove the seizure onset zone. Nine drug-responsive patients (61 remaining) and fifteen drug-resistant patients (47 remaining), whose EEG recordings exhibited PSWE activity in 99% of the duration, were also excluded from the study. On average, each patient exhibited 25.60 ± 17.60 PSWE events.

5.2 Features distribution

5.2.1 TUH data

Consistent with previous research, the total time spent in PSWE was significantly higher in epilepsy patients compared to non-epilepsy patients ($p=0.0001$). Additionally, significant differences were observed in the mean duration of PSWEs and the mean MPF per event. However, no significant differences were found between the groups in the number of PSWEs per minute, the normalized energy of PSWEs, or the mean number of channels involved in PSWEs (Figure 5.1).

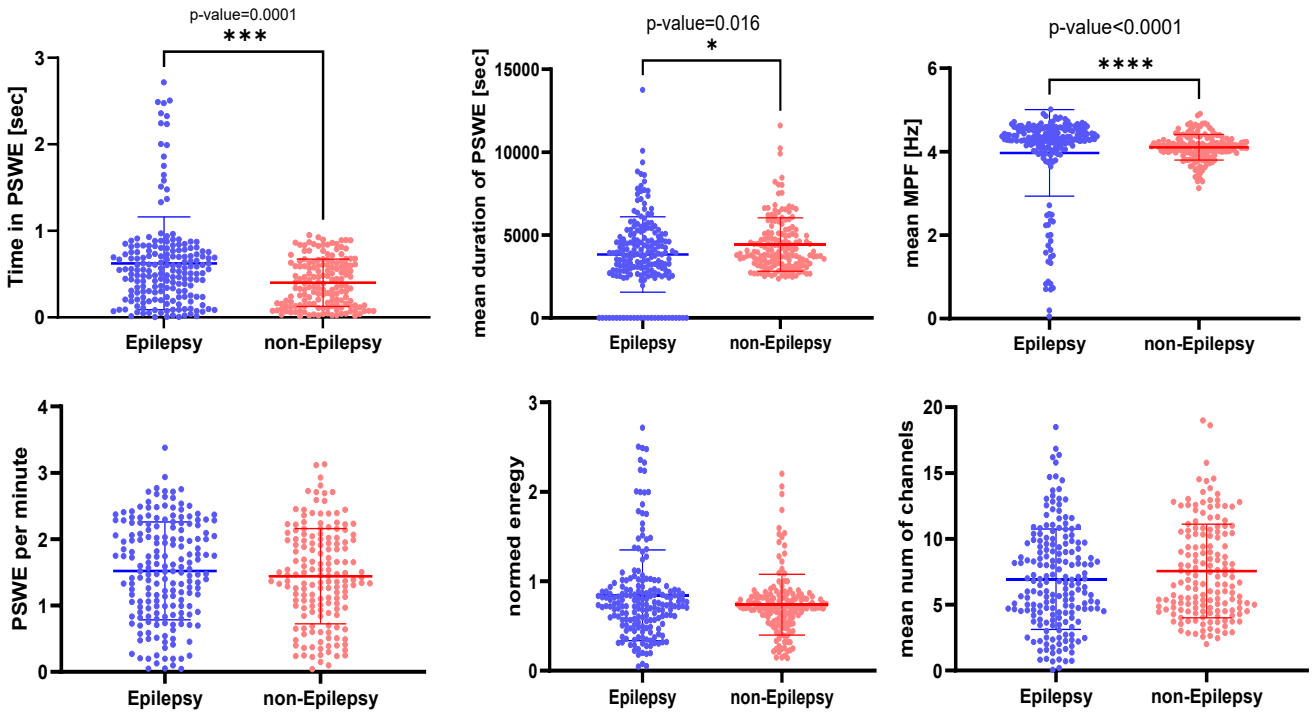


Figure 5.1 Statistical analysis of PSWE differences in patient groups

Additionally, we conducted multiple t-tests to determine whether there were significant differences between the two groups (epilepsy patients and non-epileptic individuals). The Family-Wise Error Rate (FWER) is the probability of making at least one Type I error (a false positive, where a true null hypothesis is incorrectly rejected) among a family of tests. The FWER is calculated as:

$$FWER |_{\alpha=0.05} = 1 - 0.95^m ; m \text{ is number of tests.}$$

Here, $m=39$, as 39 features were analyzed. Therefore, the probability of making at least one false positive is: $FWER |_{\alpha=0.05} = 1 - 0.95^{39} = 0.865$. To address this, we applied the Bonferroni Correction, which aims to ensure that there is less than a 5% chance of any false positive across all 39 tests. The adjusted significance level is calculated as:

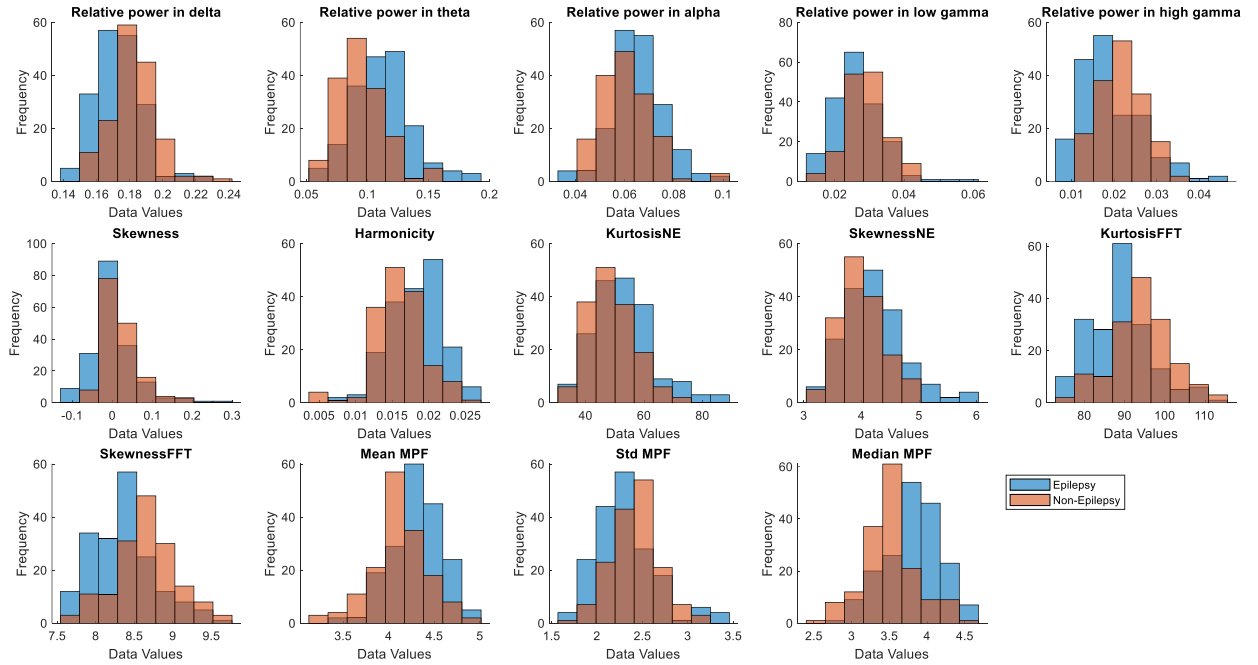
$$\alpha_{adjusted} = \frac{\alpha}{m - i + 1} ; \text{ where } i \text{ is the rank of the } p \\ - \text{ value (sorted from smallest to largest)}$$

Our analysis revealed that 14 features were significantly different (Table 1), indicating notable differences in PSWE features between the two groups. A key finding is that relative power in the theta and alpha bands was significantly higher in patients with epilepsy compared to non-epilepsy individuals, whereas in all other frequency bands (delta, low gamma, and high gamma), relative power was significantly lower in patients with epilepsy. This suggests distinct spectral differences in PSWE activity, emphasizing a specific frequency distribution in epilepsy. These changes in frequency power across different bands may indicate disruptions in normal brain network dynamics, highlighting specific neurophysiological alterations associated with epilepsy, between seizures.

Cohen's d , a statistical measure used to quantify effect size or the standardized difference between two means, was also calculated. A medium effect size is defined as Cohen's $d > 0.5$, while a large effect size is considered when Cohen's $d > 0.8$. Table 1 shows the features that differed significantly between groups, including Harmonicity, FFT kurtosis, FFT skewness, relative delta power, and relative theta power. Figure 5.2 presents the distribution of significant features.

Table 5.1: Significant T-test results

Feature na	p- value	Cohen's d
Skewness	1.55E-03	-3.46E-01
Harmonicity	4.68E-09	6.53E-01
Kurtosis NE	2.73E-05	4.62E-01
Skewness NE	1.43E-05	4.78E-01
Kurtosis FFT	2.56E-10	-7.08E-01
Skewness FFT	1.52E-10	-7.17E-01
Relative power in delta	2.51E-08	-6.20E-01
Relative power in theta	3.30E-12	7.85E-01
Relative power in alpha	2.41E-04	4.03E-01
Relative power in low gamma	4.22E-04	-3.87E-01
Relative power in high gamma	1.33E-04	-4.20E-01
MPF per seg	6.13E-10	6.92E-01
std MPF	2.06E-04	-4.07E-01
median MPF	7.68E-14	8.47E-01

**Figure 5.2 Example of significantly features distributions.**

5.2.2 BONN data

The BONN dataset was also analyzed to validate our findings from the TUH dataset and assess the generalizability of PSWE features across independent cohorts of epilepsy patients with different treatment responses. Drug-resistant patients exhibited a higher total time spent in PSWE compared to drug-responsive patients ($p\text{-value}=0.0085$). Furthermore, the normalized energy of PSWEs was lower in drug-resistant patients than in drug-responsive patients ($p\text{-value}<0.0001$; Fig. 2(b)). However, no significant differences were found between the groups in the number of PSWEs per minute, mean duration of PSWEs, mean MPF per event or the mean number of channels involved in PSWEs (Figure 5.3).

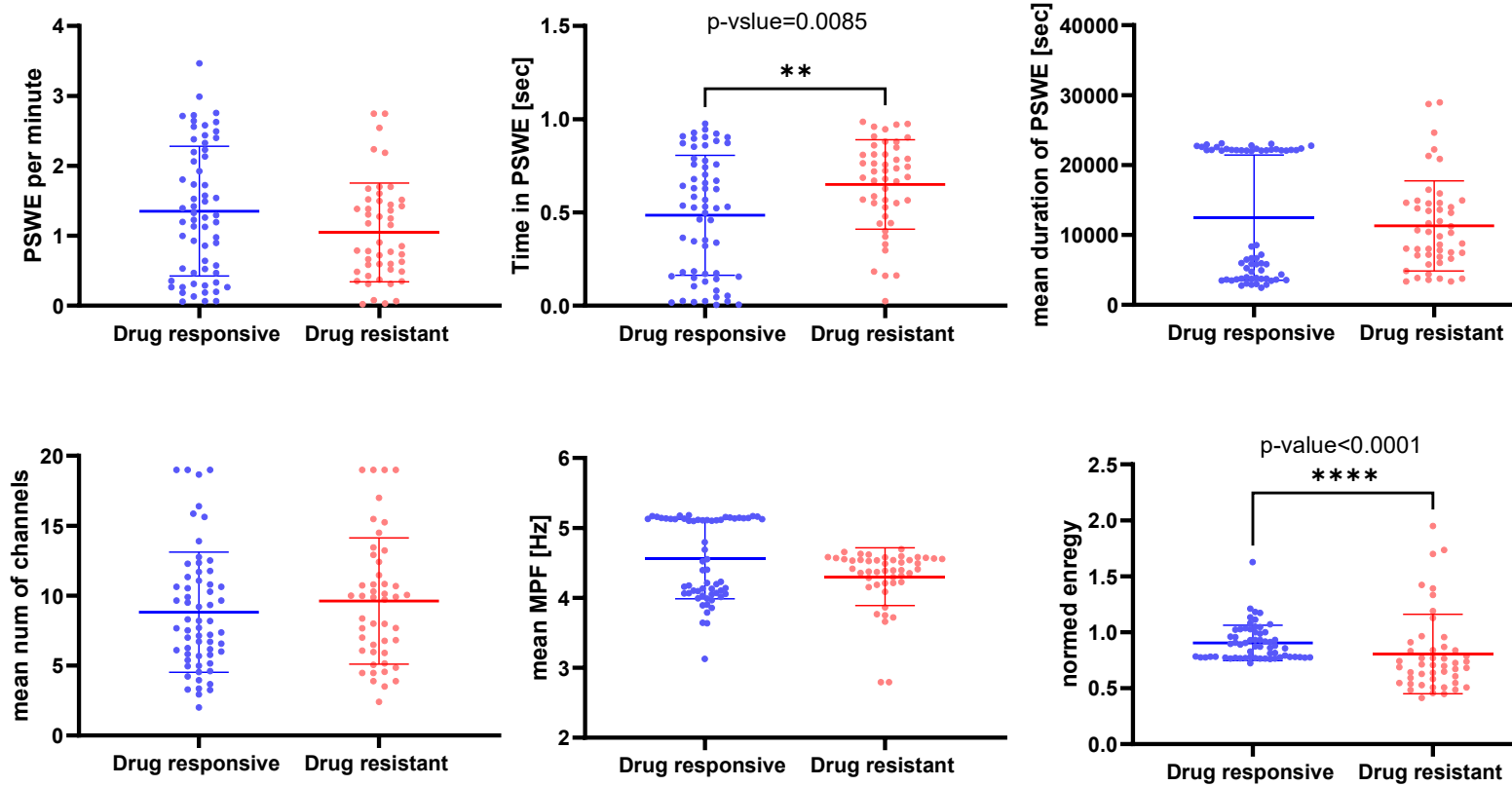


Figure 5.3 PSWE statistics in drug responsive vs. drug resistant patients

As in the analysis of TUH data, we conducted multiple t-tests with family-wise error rate (FWER) correction on the BONN dataset to identify significant differences between the two groups. Our analysis identified 10 features that were significantly different between the groups (Table 2), suggesting notable variations in PSWE characteristics. These features include kurtosis, zero crossing, kurtosis NE, relative power in all bends, standard deviation of MPF (std MPF), and high MPF. Notably,

relative power in the theta and beta bands was significantly lower in drug-responsive patients compared to drug-resistant patients. This suggests that PSWE features, particularly those related to spectral power in the theta and beta bands, could reflect underlying differences in neural dynamics between patients who respond to medication and those who do not. The distribution of these significant features is visualized in Figure 5.4, providing a detailed comparison and further supporting the observed differences in PSWE characteristics.

Table 5.2: Significant T-test results

Feature name	p- value	Cohen's d
Kurtosis	1.73E-06	-9.83E-01
Zero-crossing	2.50E-03	-6.01E-01
Kurtosis NE	2.54E-03	6.00E-01
Relative power in delta	3.72E-09	-1.25 E+00
Relative power in theta	4.43E-14	1.69 E+00
Relative power in beta	8.31E-07	1.02 E+00
Relative power in low gamma	2.90E-04	-7.28E-01
Relative power in high gamma	1.15E-07	-1.10 E+00
Std MPF	6.62E-11	-1.41 E+00
High MPF	5.50E-08	-1.41 E+00

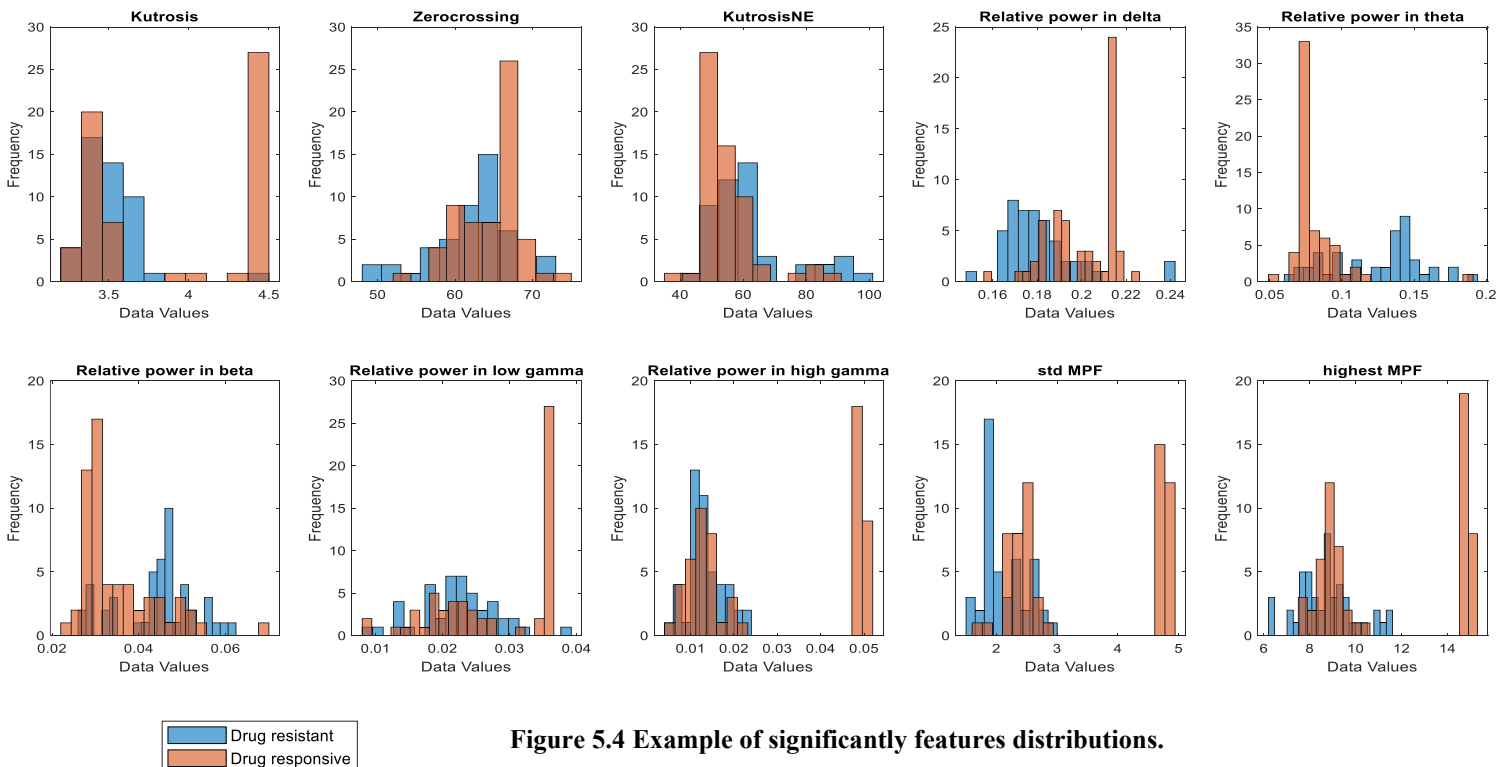


Figure 5.4 Example of significantly features distributions.

5.3 Classification per patient

5.3.1 TUH data

As detailed in the methodology section, following pre-processing, features selection was employed to identify the most optimal subsets of PSWE's features. This result focus on the classification of epilepsy and non-epilepsy based on PSWE features. To prevent overfitting and reduce the number of features, a correlation-based filter approach was applied. A threshold of 0.8 was set, meaning that features with a correlation coefficient exceeding 0.8 were excluded. Using this method, 13 features were removed including curve length, maximal amplitude, skewness NE, fractal dimension, median MPF, skewness FFT, kurtosis FFT, harmonicity, mean peak, number of PSWE, mean NE, sum FFT, and STD NE. Figure 5.5 illustrates the correlation matrix prior to feature selection.

Since 26 features remain, FFS and k-fold cross validation preformed. Figure 5.6 presents an example of the area under the curve (AUC) of the ROC, chosen as the optimal criterion, plotted against the number of features included in the model. The performance of the algorithm was evaluated on the testing data, which remained unseen for all five models (K models, as K set to 5). After testing various classification techniques (SVM, decision trees, logistic regression, etc.), the 'Gaussian SVM' method yielded the best results, with a sensitivity of 78.12%, a specificity of 78.38%, and an overall accuracy of 78.26% (Figure 5.7). The algorithm was also re-evaluated using different correlation thresholds; however, the results were less favorable. Additionally, we attempted to use only the significant features identified in the prior t-test, but this approach did not improve the results.

5.3.2 BONN data

Since all BONN patients included in this study are diagnosed with epilepsy (132 patients, with 108 not identified as persistently slow), we evaluated the performance of our previously developed classification model. This model was trained on TUH data to differentiate between epilepsy and non-epilepsy patients. Using Gaussian SVM and logistic regression models, the classifier demonstrated strong performance, correctly identifying 87.04% (94/108) and 95.37% (103/108) of epilepsy cases, respectively. These findings reinforce the reliability and robustness of our prior algorithm in accurately classifying unseen data from a different patient population. Additionally, this analysis provides insight into the relevance of PSWE features used in the models. The results suggest that these features are not only effective for distinguishing epilepsy patients but are also commonly observed in both drug-resistant and drug-responsive epilepsy cases.

Next, we trained a new model based on PSWE features to classify between drug-resistant and drug-responsive patients. Feature selection algorithms based on correlation and forward feature selection (FFS), as previously discussed, were applied. Using the correlation-based filter approach, six features were removed: relative power in the low gamma band, standard deviation of the mean peak frequency (STD MPF), kurtosis, high mean peak frequency (high MPF), standard deviation (STD), and relative power in the theta band. With 33 features remaining, FFS and k-fold cross-validation were performed. After testing various classification techniques (SVM, decision trees, logistic regression, etc.), the logistic regression method yielded sensitivity of 88.89%, specificity of 93.33%, and an overall accuracy of 91.67% (Figure 5.8).

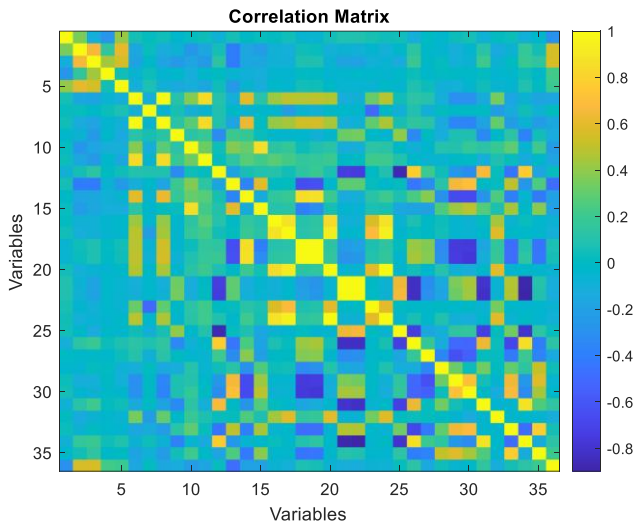


Figure 5.5 Correlation matrix of 36 features.

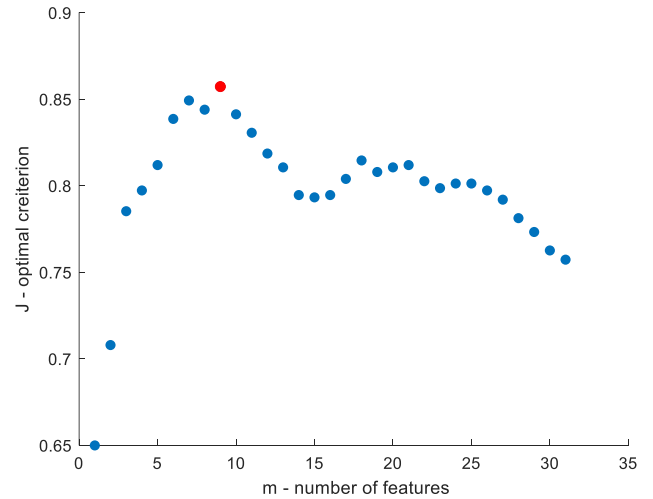


Figure 5.6 Example of Forward feature selection algorithm

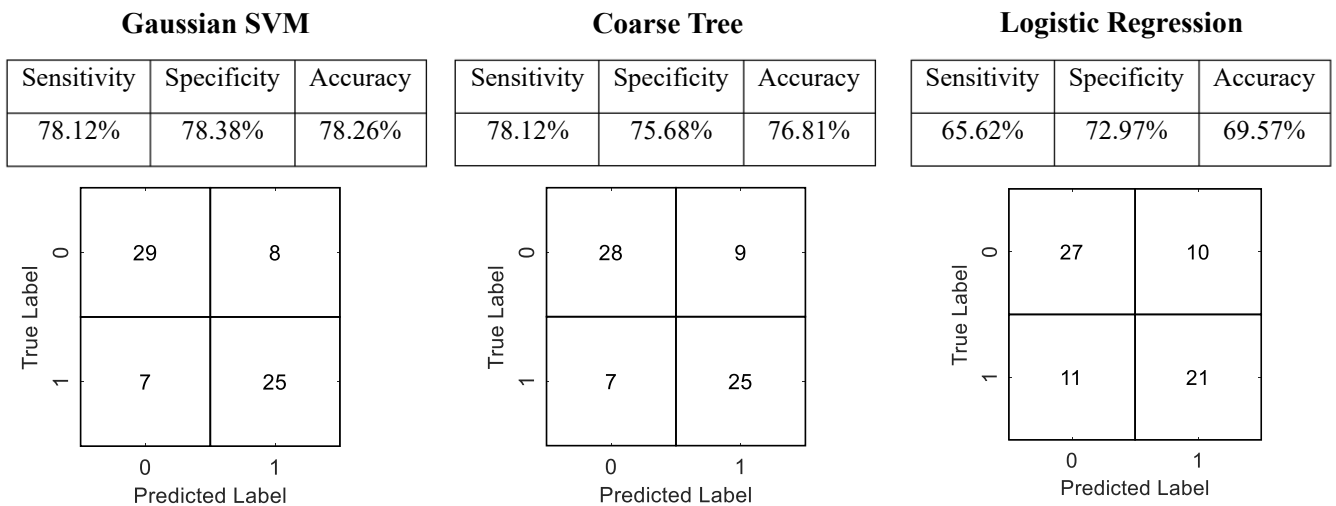


Figure 5.7 Classification models performance in TUH dataset: 1 – Epilepsy, 0 – non-Epilepsy

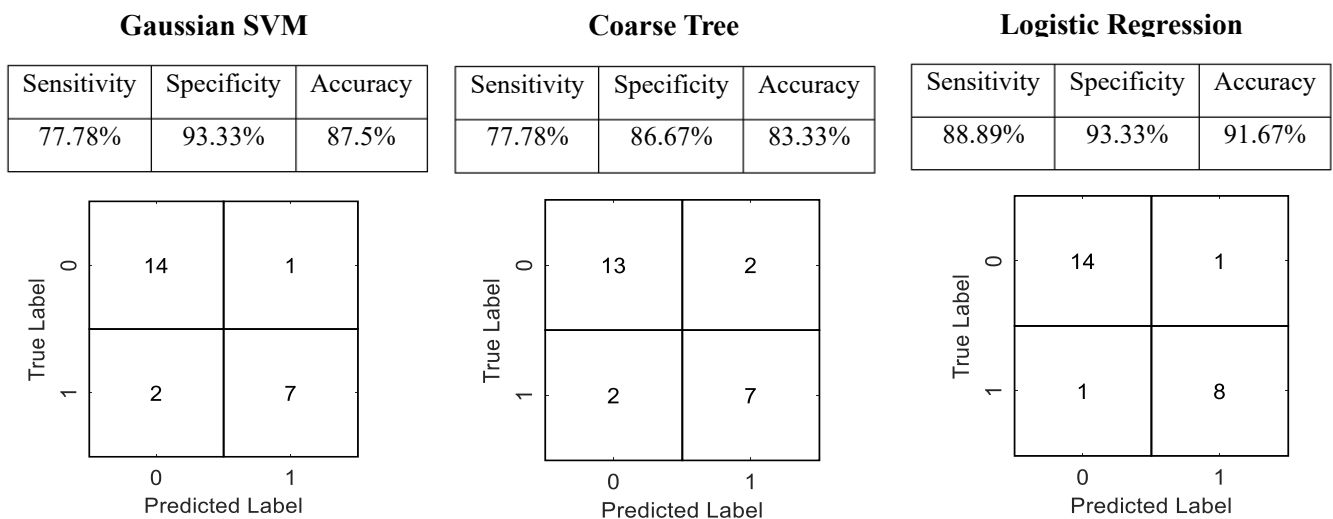


Figure 5.8 Classification models performance in BONN dataset: 1 – Drug resistant, 0 – Drug responsive.

5.4 Classification per event

5.4.1 TUH data

Using the previous coarse tree classification model, we aimed to identify pathological events based on feature patterns, hypothesizing that such events would occur more frequently in epilepsy patients. This assumption suggested that epilepsy patients might exhibit distinct and consistent feature patterns. After classification, we compared the distribution of pathological events between epilepsy and non-epilepsy groups. However, as shown in Figure 5.9, the results did not support our hypothesis; no significant differences in the percentage of time spent in pathological events were observed between the two groups.

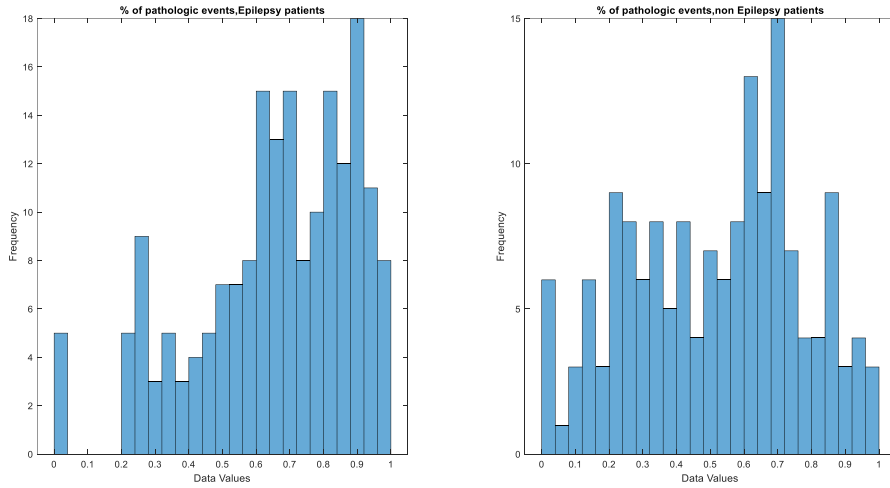
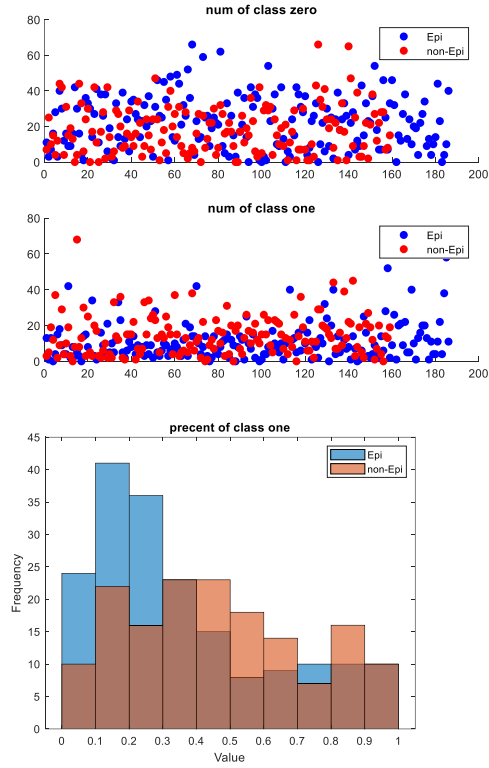


Figure 5.9 Pathologic features distribution

To further investigate our hypothesis, we applied two different unsupervised algorithms. First, we used the K-means clustering algorithm on the event features, setting $K=2$ based on the assumption that events could be classified into two categories: physiological and pathological. Figure 5.10A illustrates the distribution of these two event classes within the epilepsy and non-epilepsy groups. However, the results did not reveal any notable patterns or distinctions.

To refine the analysis, we then trained the model exclusively on the epilepsy group. This approach was based on the assumption that both types of events—physiological and pathological—would likely occur within the epilepsy group. In contrast, we could not confirm whether pathological events exist in the non-epilepsy group. Nevertheless, the results again failed to reveal any clear or notable distinctions (Figure 5.10B).

A) Classification on all the data



B) Classification only epilepsy patients

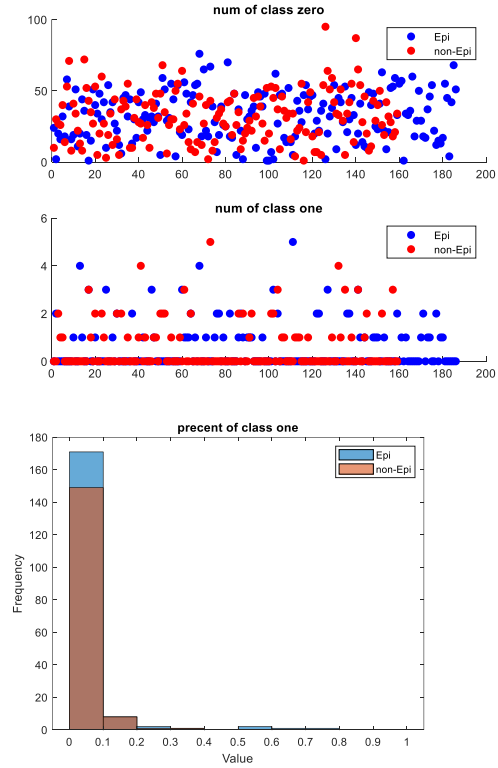


Figure 5.10 K-means results

Secondly, we applied PCA followed by a Gaussian GMM to uncover distinct patterns in the data and explore alternative classification approaches. PCA was used to reduce the dimensionality of the event features, enabling us to focus on the most significant components that capture the majority of the data's variance. We retained the first three principal components, as each explained more than 10% of the variance (25.38%, 15.81%, and 12.70%, respectively). The GMM was then applied to these principal components to model the data and classify events based on probabilistic distributions. Figure 5.11 illustrates the classification between two types of events.

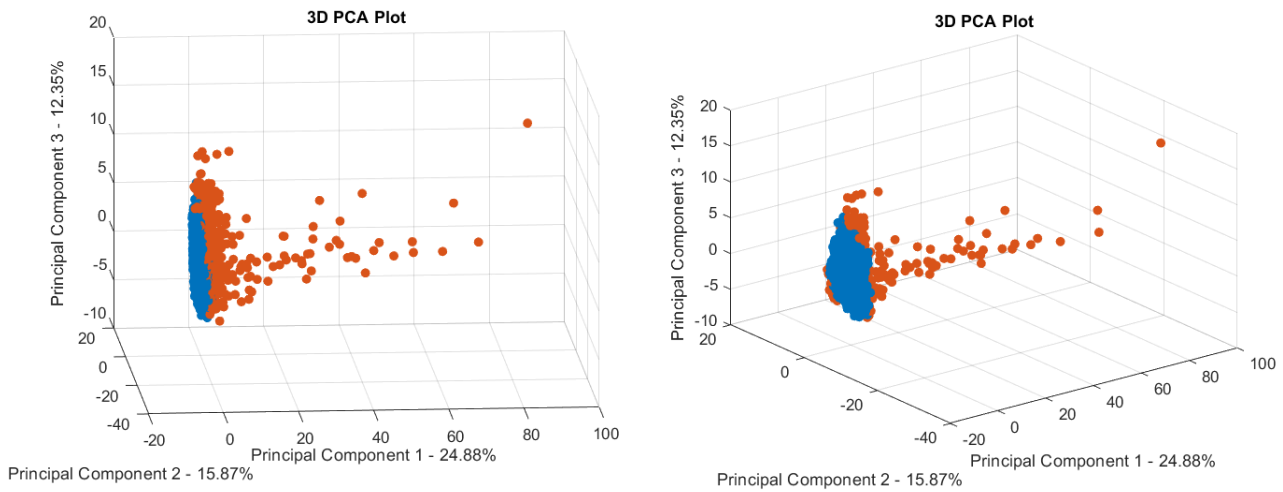


Figure 5.11 Classification of event types using the PCA+GMM algorithm.

As shown in this Figure 5.11, it appears that the "blue" type of event occurs more frequently, while the "red" type is relatively rare. Next, we examined the distribution of both event types across our patient group (Figure 5.12). The results again failed to reveal any clear or notable distinctions between epilepsy and non-epilepsy patients.

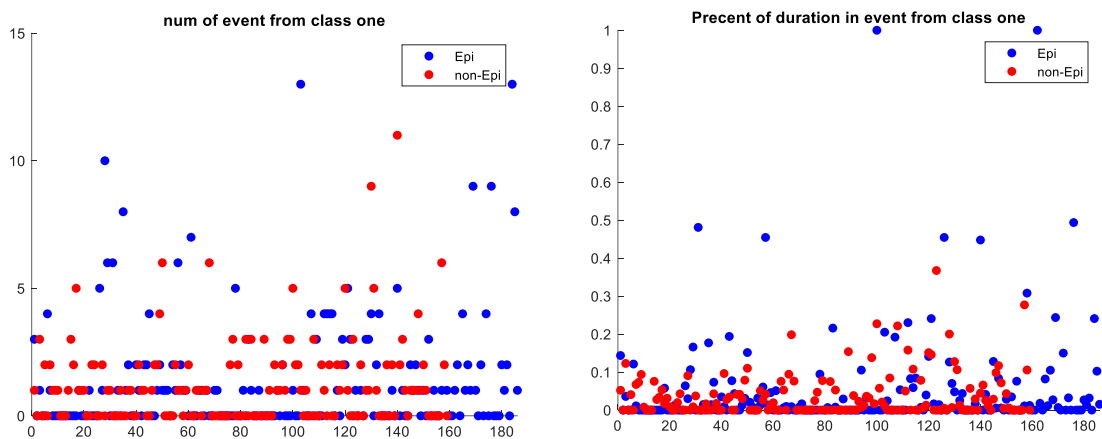


Figure 5.12 Class 1 distribution – PCA + GMM algorithm.

5.5 Spatial Analysis

In this analysis, I aimed to identify differences in the spatial distribution of PSWEs across patients' electrodes. As detailed in the methodology section, I generated topographic maps illustrating the weighted mean energy of PSWEs within each patient's electrode (Figure 4.6). First, I examined group differences in the percentage of scalp areas exhibiting PSWE activity and found significant differences in both patient groups. As shown in Figure 5.13, epilepsy patients exhibited a significantly higher percentage of affected areas compared to non-epilepsy patients ($p=0.0262$; Mann–Whitney U test). A similar comparison in the BONN dataset revealed that drug-resistant patients had a significantly higher percentage of affected areas involved than drug-responsive patients ($p=0.0002$; Mann–Whitney U test). These findings suggest that the epilepsy is associated with a broader spatial distribution of PSWEs in the brain.

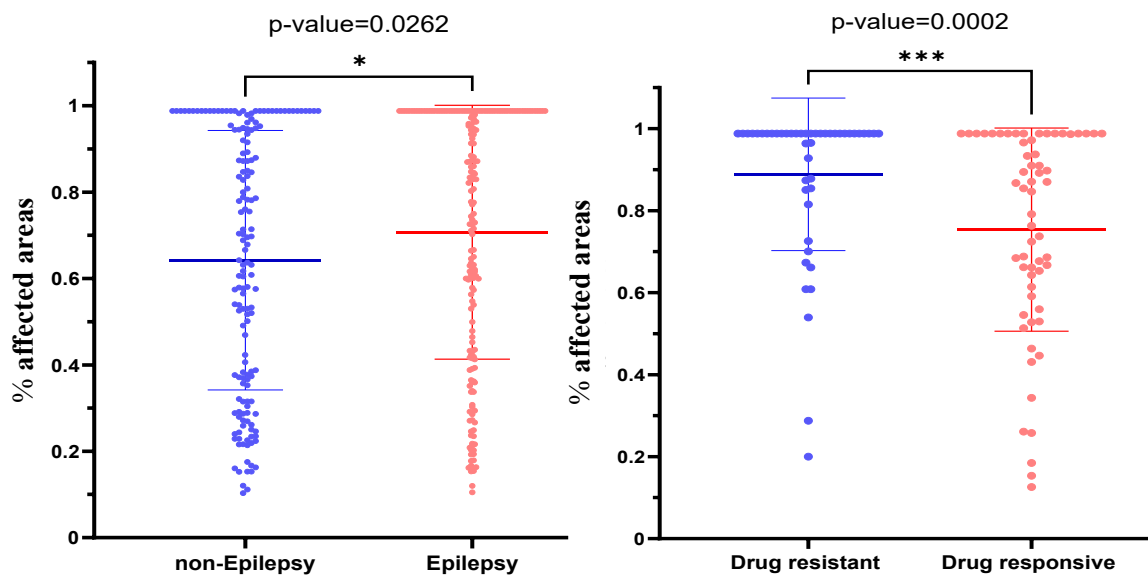


Figure 5.13 Percent of affected areas in PSWEs from the topographic map.

To further support this hypothesis, I applied image processing algorithms to segment the scalp into distinct regions and classified each patient based on the number of regions activated during PSWE events. As shown in Figure 5.14, epilepsy patients exhibited more widespread activity, with a significantly higher proportion of patients classified as 'Multi-Focal' ($p=0.0019$ and $p=0.0005$; Fisher's exact test). Consistent with this finding, in the BONN dataset, drug-resistant patients exhibited significantly more widespread activity than drug-responsive patients, with a higher proportion classified as 'Multi-Focal' ($p=0.01456$; Fisher's exact test).

I then integrated the three measures used for clustering (WMED, DCI, and 1-D) into a single focality score, the **General Focality Index (GFI)**:

$$GFI = WMED \cdot DCI \cdot (1 - D) \quad (5.1)$$

A high GFI indicates the presence of a single, compact, and energetically dominant region, whereas a low GFI reflects distributed activity without a clear dominant focus:

- Low DCI: no dominant cluster by size
- Low WMED: no significantly dominant cluster by energy
- High D: cluster centers are far apart

Using this metric, I found that the GFI, which quantifies the degree of focality in brain activity by integrating spatial clustering, energy dominance, and dispersion of PSWE regions, was significantly lower in epilepsy patients compared to non-epilepsy patients (Figure 5.15, $p=0.0066$; Mann-Whitney U test) compared to non-epilepsy patients, indicating less focal activity. Furthermore, drug-resistant patients exhibited significantly lower GFI than drug-responsive patients (Figure 5.15, $p=0.0264$; Mann-Whitney U test). These results highlight a clear association between the spatial extent of PSWE activity and disease severity. The higher prevalence of 'Multi-Focal' patterns and lower GFI values among epilepsy and drug-resistant patients suggest that PSWE events in these groups are not confined to a single localized region, but instead involve multiple, spatially distinct brain areas. The findings support the hypothesis that increased spatial dispersion of PSWEs is a hallmark of more severe or treatment-resistant forms of epilepsy and may serve as a potential biomarker for disease progression or response to therapy.

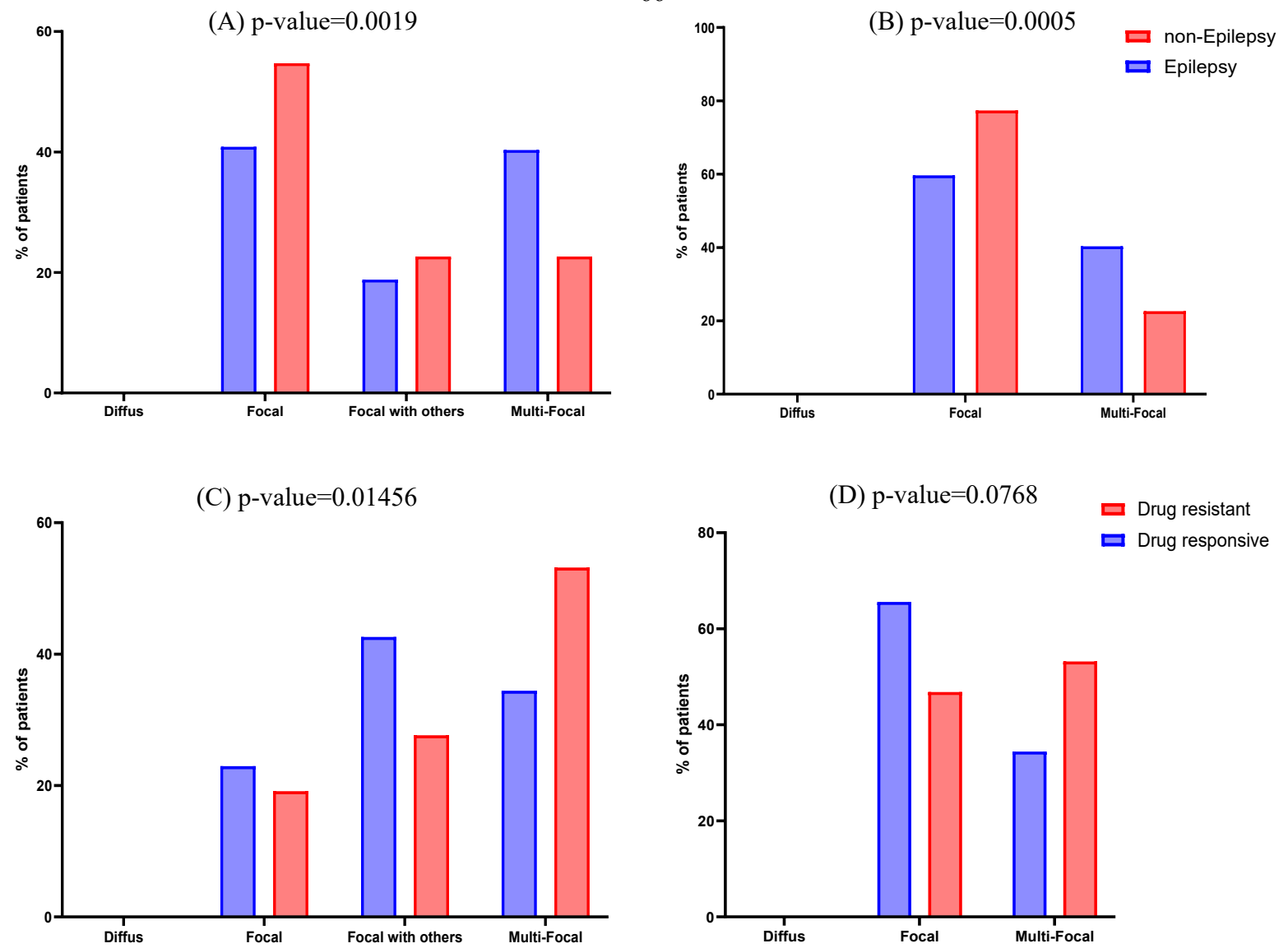


Figure 5.14 Differences in PSWE spatial clusters between patient groups. (A-B) TUH datasets; (C-D) BONN dataset. (A, C) Show the percentage of each cluster type: 'Focal', 'Focal with other centers', and 'Multi-focal'. (B, D) Present the same data with 'Focal' and 'Focal with other centers' combined into a single category.

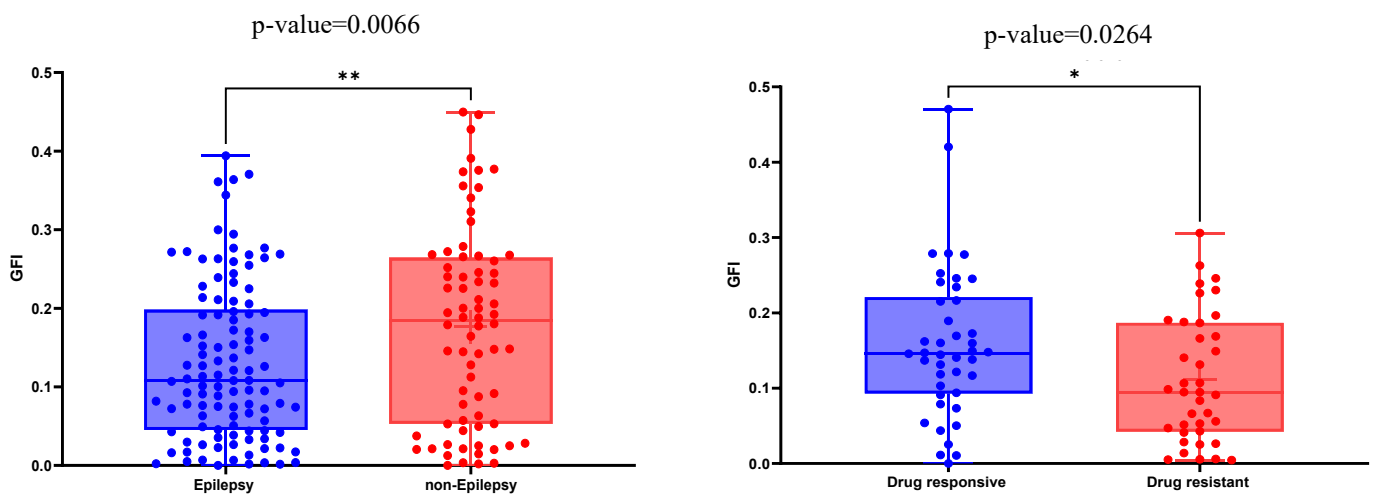


Figure 5.15 GFI significant differences. Left: TUH datasets; Right: BONN dataset.

6 Discussion

Epilepsy is one of the most prevalent neurological disorders worldwide, often leading to a reduced quality of life and cognitive comorbidities. Despite advances in diagnostic tools and treatment strategies, there remains a critical need for reliable biomarkers that can aid in both the early diagnosis and prediction of epilepsy, particularly in cases where seizures are infrequent or difficult to capture with standard EEG recordings. In recent studies, paroxysmal slow wave events (PSWEs) have emerged as promising EEG-based biomarkers, not only for epilepsy but also for a range of neurological and neurodegenerative conditions, including Alzheimer's disease, Parkinson's disease, obstructive sleep apnea, and aging-related cognitive decline. These events have been associated with pathological slowing of brain activity and linked to processes such as blood-brain barrier disruption and disease progression. However, despite their potential, previous studies have largely focused on the presence, frequency, and duration of PSWEs, with limited attention to their spatial and temporal characteristics or how these features may differ across clinical populations. The lack of comprehensive, comparative analyses has hindered a deeper understanding of the pathophysiological relevance of PSWEs and their diagnostic utility. The present study addresses this gap by systematically characterizing PSWEs across both healthy and diseased brains, with a particular focus on epilepsy. I examined the spatial, temporal and spectral characteristics of PSWEs across two independent EEG datasets- TUH and BONN, in order to identify robust biomarkers associated with epilepsy presence and severity. The results reveal several consistent and significant group-level differences that provide insights into the neurophysiological alterations accompanying epilepsy.

In the TUH dataset, epilepsy patients demonstrated a significantly higher total time spent in PSWE episodes compared to non-epileptic controls. Importantly, detailed spectral analysis revealed a distinct shift in frequency power: patients with epilepsy showed increased relative power in the theta (3-8 Hz) and alpha (8-12 Hz) bands, but decreased power in the delta, low gamma, and high gamma bands. In addition, an important limitation of the present work is related to the composition of the TUH dataset: the 'non-epilepsy' group includes patients who were only not known to have epilepsy at the time of recording, which does not exclude undiagnosed cases and may introduce heterogeneity into the control population.

In the BONN dataset, we observed a parallel pattern: drug-resistant epilepsy patients exhibited longer cumulative time in PSWE and lower normalized energy per event compared to drug-responsive patients. Notably, spectral differences also emerged: drug-resistant patients showed higher relative power in theta and beta bands, indicating possible compensatory mechanisms or altered excitability patterns.

When comparing the TUH and BONN datasets, several consistent and divergent patterns emerged. In both cohorts, the total time spent in PSWE was significantly higher in patients with epilepsy (TUH: epilepsy vs. non-epilepsy; BONN: drug-resistant vs. drug-responsive), highlighting this feature as a robust and reproducible marker of pathological slowing across different clinical contexts. Spectral differences, however, showed partial divergence: TUH epilepsy patients exhibited increased relative power in the theta and alpha bands (and decreased power in delta and gamma), whereas BONN drug-resistant patients were characterized by elevated theta and beta power. This suggests that while PSWEs consistently capture abnormal slowing, the specific frequency shifts may reflect cohort-specific dynamics. Additional features such as kurtosis and MPF-related measures were significant in both datasets, indicating stable discriminative value, whereas others (e.g., harmonicity in TUH, zero-crossing in BONN) were dataset-specific. Overall, these findings demonstrate that PSWEs embody both universal markers of epileptic brain activity and population-dependent variations, underscoring the importance of cross-dataset validation.

After showing that there is a significant difference between patient groups in PSWE features (14 significant features in the TUH dataset and 10 in the BONN dataset), I used PSWE features in order to train a classification model. The results of the classification models further underscore the robustness of PSWE-based features. Using Gaussian SVMs and logistic regression models trained on the TUH data, we achieved over 78% accuracy in distinguishing between epilepsy and non-epilepsy patients. More notably, these models generalized well to the BONN dataset, correctly classifying 87% (SVM) to 95% (logistic regression) of patients, despite the inherent heterogeneity in recording setups and patient demographics between the two datasets. This highlights the generalizability and translational potential of PSWE features in broader clinical contexts. A secondary classification model trained specifically on the BONN data to differentiate between drug-resistant and drug-responsive patients yielded a high

classification accuracy of 91.67%, demonstrating the potential of PSWE features in stratifying epilepsy patients by treatment responsiveness.

Also, the spatial analysis revealed compelling group-level differences in the distribution of PSWE activity across the cortex. Epilepsy patients, as well as drug-resistant patients, exhibited a greater percentage of brain regions engaged during PSWE episodes, supporting the hypothesis that epilepsy is associated with broader network-level dysfunction. Additionally, the proportion of patients classified as 'Multi-Focal' based on the number of activated regions was significantly higher among the epilepsy and drug-resistant groups, reinforcing the link between spatial dispersion of PSWEs and disease severity.

Taking together, these findings strongly suggest that PSWEs are not random slow wave phenomena, but instead reflect underlying pathological processes related to epilepsy. The consistency of results across datasets and analysis levels underscores their potential as non-invasive, EEG-based biomarkers for both epilepsy diagnosis and stratification. In particular, classification models trained on PSWE features achieved over 78% accuracy in distinguishing epilepsy from non-epilepsy patients (TUH dataset) and 91.67% accuracy in differentiating drug-resistant from drug-responsive patients (BONN dataset), further underscoring the robustness and clinical relevance of PSWE-based features. Moreover, epilepsy and drug-resistant patients exhibit more widespread, multi-focal PSWE activity, indicating that increased spatial dispersion of PSWEs is linked to greater disease severity and treatment resistance. These findings highlight PSWE characteristics as reliable biomarkers that capture both disease presence and severity, with strong implications for patient monitoring and therapeutic decision-making. Future studies should explore temporal and spatial changes in PSWE features over time, and leverage larger datasets combined with deep learning approaches to improve classification performance and clinical applicability.

7 Conclusion

This study provides strong evidence that paroxysmal slow wave events (PSWEs) capture meaningful neurophysiological alterations in epilepsy and represent promising EEG-based biomarkers. Across two independent datasets, PSWE characteristics consistently distinguished between patients with and without epilepsy and further stratified drug-resistant from drug-responsive patients. Using features extracted from PSWEs, classification models trained on the TUH dataset successfully distinguished epilepsy from non-epilepsy patients, while models trained on BONN data stratified drug-resistant and drug-responsive patients. These findings indicate that PSWE-based features can not only support early diagnosis and prediction of epilepsy, but also assist in personalized treatment planning by identifying patients likely to respond to specific therapies. Additionally, spatial analysis revealed greater cortical involvement and multi-focal PSWE patterns in epilepsy and drug-resistant patients, suggesting that the spatial distribution of PSWEs may serve as a marker of disease severity and network-level dysfunction. The reproducibility of PSWE features across independent datasets underscores their translational potential as non-invasive biomarkers. Future research should aim to validate these findings in larger, more diverse populations and in longitudinal studies. In addition, integrating PSWE features with advanced computational approaches, such as deep learning, could further improve predictive accuracy and enable the development of automated clinical tools. Exploring the temporal changes of PSWEs and their relationship to treatment response or disease progression will provide deeper insights into their role as dynamic biomarkers.

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תקציר

האטה בפעילות המוחית הנצפית ברישומי אלקטרואנצפלוגרם (EEG) היא תופעה שכיחה בתנאי מנוחה, כגון שינה [1]. עם זאת, מחקרים זיהו דפוס מובחן של האטה קורטיקלית אצל חולים עם הפרעות נוירולוגיות, כגון אפילפסיה ומחלת אלצהיימר [2], [3], המתבטא באירועי גל איטי התקפי (paroxysmal slow wave events – PSWEs). אירועים אלו מוגדרים כגלים בעלי תדר חציוני נמוך מ-6 הרץ ובאורך של יותר מ-5 שניות. במחקר זה ניתחתי רישומי EEG קליניים ממסדי הנתונים של Temple University ו-Bonn University במטרה לאפיין באופן שיטתי את המאפיינים הזמניים, הספקטראליים והמרחביים של PSWEs בחולי אפילפסיה, כולל תת-קבוצות של חולים עמידים לתרופות ורגישים לתרופות. לאחר חילוף מאפיינים של PSWEs אימנתי מודלים של למידת מכונה במטרה לסווג נוכחות של מחלת האפילפסיה ועמידות לתרופות על סמך המאפיינים של גלים אלו. הממצאים מצביעים על כך שתכונות PSWE מבדילות באופן מהימן בין חולי אפילפסיה לחולים בריאים, וכן מאפשרות להבחין בין חולים עמידים לתרופות לחולים רגישים לתרופות. מודלי הסיווג הראו דיוק גבוה בשני מסדי הנתונים, מה שמעיד על אמינותם ויכולת הכללה של ביו-מרקרים המבוססים על PSWE. בנוסף, ניתוח מרחבי של גלים אלו הדגים את מעורבות הקורטקס ואת הדפוסים הרב-מוקדיים של פעילות הגלים. ממצאים אלה מדגישים את PSWEs כסמנים לא-פולשניים ברישומי EEG, המספקים מידע על נוכחות המחלה וחומרתה. מחקרים עתידיים צריכים לאמת את התוצאות בקבוצות חולים גדולות יותר ובהקלטות וארוכות טווח, ולבחון שילוב עם שיטות חישוביות מתקדמות, כגון למידה עמוקה, על מנת לשפר את הזיהוי האוטומטי ולהעצים את תהליכי קבלת ההחלטות הקליניות.

מילות מפתח: אירועי גל איטי התקפי (PSWEs); אפילפסיה; ביו-מרקרים ב-EEG; אפילפסיה עמידה לתרופות; למידת מכונה; ניתוח ספקטראלי; מודלי סיווג; תכונות EEG זמניות ומרחביות.

אוניברסיטת בן-גוריון בנגב
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המחלקה להנדסה ביו רפואית

ניתוח מבוסס נתוני EEG של אירועי גל איטי התקפי בהפרעות מוחיות

חיבור זה מהווה חלק מהדרישות לקבלת תואר מגיסטר בהנדסה

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